Practice List, Lecture 3: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 18 exercises of `Listas/pratica-lecture-03.md`

November 2026

These are detailed solutions to the 18 book exercises of the list `pratica-lecture-03.md` (Lecture 3: mixed strategies, extensive form, backward induction and subgame perfection), in the list's order from easy to hard; each heading shows the item's number, level and difficulty from 1 (apply a definition) to 5 (graduate level). Each statement is printed verbatim from the list right before its solution. Every solution has the same six parts: (a) what is asked; (b) the method, with book section, page and lecture slide; (c) the solution, every step shown; (d) the economic interpretation; (e) the verification, by `check_L03.py` in this folder (new items) or the check scripts of `lista-reais-lectures-01-05-solutions/` (reused items), and against the book's worked answer where one exists; (f) common mistakes. Slide numbers in the reused solutions refer to the drafts of the earlier real-exercise list and may differ by a few slides from the final `Aula/Lecture03.pdf`. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 3 — Mixed strategies and dynamic games

Item 1: Tadelis Exercise 6.5

Foundations · difficulty 2/5

Statement (verbatim from the list, item 1)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 6.5 "Cops and Robbers", p. 124 [PDF 143]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Cops and Robbers:** Player 1 is a police officer who must decide whether to patrol the streets or to hang out at the coffee shop. His payoff from hanging out at the coffee shop is 10, while his payoff from patrolling the streets depends on whether he catches a robber, who is player 2. If the robber prowls the streets then the police officer will catch him and obtain a payoff of 20. If the robber stays in his hideaway then the officer's payoff is 0. The robber must choose between staying hidden or prowling the streets. If he stays hidden then his payoff is 0, while if he prowls the streets his payoff is −10 if the officer is patrolling the streets and 10 if the officer is at the coffee shop.
>
> a. Write down the matrix form of this game.
> b. Draw the best-response function of each player.
> c. Find the Nash equilibrium of this game. What kind of game does this game remind you of?

(a) What is asked. A police officer (player 1) chooses between patrolling the streets and the coffee shop; a robber (player 2) chooses between staying hidden and prowling. (a) Write the matrix; (b) draw each player's best-response function; (c) find the Nash equilibrium and say which game it recalls.

(b) Method. Build the matrix from the story, mark best replies, and, because every cell is a best reply of at most one player, look for an equilibrium in mixed strategies with the principle of indifference (Lecture 3, slides 21–26 and the battle of the sexes on slides 27–34; Tadelis §6.2, pp. 108–114).

(c) Solution. **(a) Matrix.** Let P = patrol, C = coffee shop for player 1 and H = hidden, S = prowl the streets for player 2. Coffee shop pays the officer 10 whatever the robber does; patrolling pays 20 if the robber prowls and 0 if he hides. The robber gets 0 hidden; if he prowls he gets −10 against a patrol and 10 against the coffee shop.

	H	S
P	$0,\ \underline{0}$	$\underline{20},\ -10$
C	$\underline{10},\ 0$	$10,\ \underline{10}$

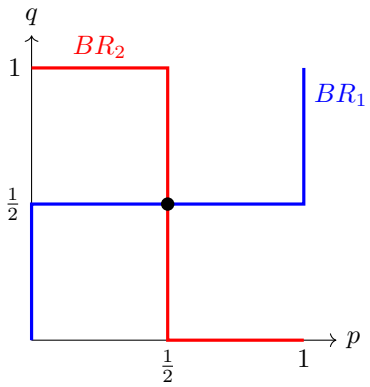
Underlined payoffs are best replies. No cell has both payoffs underlined, so there is no pure Nash equilibrium: the replies chase each other around the matrix. Against H the officer prefers C, against C the robber prefers S, against S the officer prefers P, against P the robber prefers H.

(b) Best-response functions. Let $p = \Pr(P)$ and $q = \Pr(S)$. Expected payoffs:

$$u_1(P,q) = 20q, \qquad u_1(C,q) = 10,$$
$$u_2(H,p) = 0, \qquad u_2(S,p) = -10p + 10(1-p) = 10 - 20p.$$

The officer prefers P iff $20q > 10$, i.e. $q > \frac{1}{2}$; the robber prefers S iff $10 - 20p > 0$, i.e. $p < \frac{1}{2}$. Hence

$$BR_1(q) = \begin{cases} p = 0 & q < \frac{1}{2} \\ p \in [0,1] & q = \frac{1}{2} \\ p = 1 & q > \frac{1}{2} \end{cases}, \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2} \\ q \in [0,1] & p = \frac{1}{2} \\ q = 0 & p > \frac{1}{2} \end{cases}.$$



The two graphs cross once, at $(p,q) = (\frac{1}{2}, \frac{1}{2})$.

(c) Nash equilibrium. Indifference of the robber: $0 = 10 - 20p$ gives $p = \frac{1}{2}$; indifference of the officer: $20q = 10$ gives $q = \frac{1}{2}$. The unique Nash equilibrium is

$$\sigma^* = (\tfrac{1}{2}P + \tfrac{1}{2}C, \quad \tfrac{1}{2}H + \tfrac{1}{2}S),$$

with expected payoffs $u_1 = 10$ and $u_2 = 0$. It is unique because no pure strategy of either player is a best reply to a pure strategy of the other, and a mixed equilibrium with one player pure is ruled out by the best-reply functions above (a pure p makes the robber strictly prefer one pure action, which makes the officer strictly prefer one pure action, which breaks the first).

The game recalls *matching pennies* (and the penalty kick of slide 3): one player wants to match the other's action, the other wants to mismatch, so each must be unpredictable. It is an inspection game rather than a zero-sum game (payoffs do not sum to a constant), but the logic is the same.

(d) Interpretation. The equilibrium probabilities are set by the *other* player's payoffs: the officer patrols half the time because that is what makes the robber indifferent, not because patrolling is attractive in itself. If the robber's fine for being caught grew from 10 to 20 (payoff -20 against a patrol), $u_2(S,p) = 10 - 30p$ and the officer would patrol only a third of the time; the officer itself is unaffected in equilibrium (still 10).

(e) Verification. `check_L03.py` finds no pure equilibrium, solves both indifference equations exactly, checks that $(\frac{1}{2}, \frac{1}{2})$ is a Nash equilibrium against all pure deviations, finds it is the only mixed equilibrium by support enumeration, and recomputes the patrol probability $\frac{1}{3}$ in the larger-fine variant. Output line: `tadelis-6-5: no pure NE; mixed NE (1/2,1/2), payoffs (10, 0)`. Tadelis prints no answer.

(f) Common mistakes.

- Computing the officer's mix from the *officer's* indifference: it makes the officer indifferent, not the robber, and gives q, the robber's probability.
- Reporting the equilibrium payoff of the officer as $20 \cdot \frac{1}{2} = 10$ from patrolling only, or forgetting that the robber's expected payoff is exactly 0 (his payoff from hiding).
- Stopping at "no equilibrium": there is no equilibrium *in pure strategies*; Nash's theorem gives one in mixed strategies.

Answer. (a) The matrix above (P, H: $(0, 0)$; P, S: $(20, -10)$; C, H: $(10, 0)$; C, S: $(10, 10)$). (b) BR_1 switches at $q = \frac{1}{2}$, BR_2 at $p = \frac{1}{2}$. (c) No pure equilibrium; the unique Nash equilibrium is mixed, $p = \Pr(\text{patrol}) = \frac{1}{2}$, $q = \Pr(\text{prowl}) = \frac{1}{2}$, with payoffs $(10, 0)$; the game resembles matching pennies.

Item 2: Tadelis Exercise 7.1

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.1 "Strategies", p. 147 [PDF 166]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Strategies:** Imagine an extensive-form game in which player i has K information sets.
>
> a. If the player has an identical number of m possible actions in each information set, how many pure strategies does he have?
>
> b. If the player has m_k actions in information set $k \in \{1, 2, \ldots, K\}$, how many pure strategies does the player have?

(a) What is asked. Count player i's pure strategies when he has K information sets with (a) m actions at each, (b) m_k actions at information set k.

(b) Method. A pure strategy is a complete plan: one action at *every* information set, including those the plan itself never reaches (Lecture 3, slide 25; Tadelis §7.2.1, Def. 7.4, pp. 137–138). So the set of pure strategies is the Cartesian product of the action sets, $S_i = \times_{h_i \in H_i} A_i(h_i)$, and its size is the product of their sizes (counting principle).

(c) Solution. **(a)** Label player i's information sets $h^1, \ldots, h^K$. A pure strategy is a list $s_i = (s_i(h^1), \ldots, s_i(h^K))$ with $s_i(h^k) \in A_i(h^k)$ and $|A_i(h^k)| = m$. The choice at h^1 can be made in m ways; for each of them the choice at h^2 in m ways, and so on:

$$|S_i| = \underbrace{m \cdot m \cdots m}_{K \text{ times}} = m^K.$$

(b) The same argument with $|A_i(h^k)| = m_k$:

$$|S_i| = m_1 \cdot m_2 \cdots m_K = \prod_{k=1}^{K} m_k.$$

Examples. In the sequential battle of the sexes (Lecture 3, slide 26) the follower has $K = 2$ information sets with $m = 2$ actions, so $2^2 = 4$ strategies (OO, OF, FO, FF). A player with information sets of sizes $(2, 3, 4)$ has $2 \cdot 3 \cdot 4 = 24$ pure strategies.

Mixed versus behavioral strategies. The count also measures how large the two kinds of randomization are (Lecture 3, slide 27; Tadelis §7.2.2, Defs. 7.5–7.6, pp. 139–140). A mixed strategy is a point in the simplex over $\prod_k m_k$ pure strategies, which has dimension $\prod_k m_k - 1$; a behavioral strategy picks one distribution per information set, dimension $\sum_k (m_k - 1)$. With $(m_k) = (2, 3, 4)$ these are 23 and 6. Under perfect recall the smaller space already generates every distribution over outcomes (Kuhn's theorem, Lecture 3, slide 28; Tadelis p. 142).

(d) Interpretation. The count grows exponentially in the number of information sets: what makes a dynamic game "big" is the number of contingencies, not the number of moves actually played. This is why the normal form of even a small tree (the Brexit game of Lecture 4, slides 7–8, gives the EU nine strategies) is cumbersome, and why backward induction, which works node by node, is the practical tool.

(e) Verification. `check_B_L03.py` enumerates every plan with `itertools.product` for $(m_k) = (2,2),(3,2,2),(2,3,4),(5)$ and for $m = 3$, $K = 4$, and compares the count with the formulas; it also checks the two dimensions 23 and 6. Output line: `tadelis-7-1: m^K and prod m_k confirmed by enumeration`. Tadelis prints no answer.

(f) Common mistakes.

- Adding instead of multiplying (Km or $\sum_k m_k$): that counts *actions*, which is the dimension count of behavioral strategies plus K, not the number of plans.
- Leaving out information sets that a plan makes unreachable. A strategy specifies an action there too (the off-path parts are exactly what subgame perfection tests).
- Counting decision nodes instead of information sets: nodes in the same information set share one action choice.

Answer. (a) m^K. (b) $\prod_{k=1}^{K} m_k$.

Item 3: Muñoz-Garcia Ch. 4, Exercise 3

Foundations · difficulty 1/5

Statement (verbatim from the list, item 3)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 3 "Sequential Version of the Prisoner's Dilemma Game", p. 160 [PDF 172]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider the game tree in Fig. 4.9, representing the sequential-move version of the Prisoner's dilemma game, where player 1 acts first, and observing its move player 2 responds.

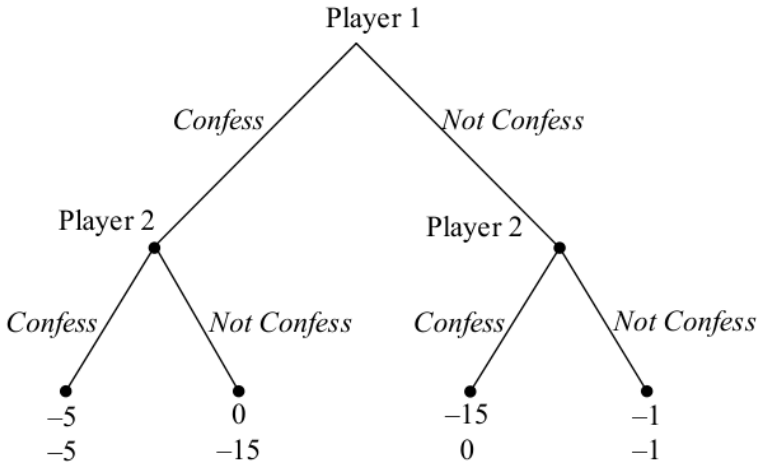


Fig. 4.9 Prisoner's dilemma sequential-move game

(a) Find the best responses of the second mover, for each of the first mover's actions.

(b) Find the first mover's equilibrium action and describe the subgame perfect Nash equilibrium of the game.

(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

(a) What is asked. In the sequential prisoner's dilemma of Fig. 4.9: (a) player 2's best reply to each move of player 1; (b) player 1's move and the subgame-perfect equilibrium; (c) whether play differs from the simultaneous version (Ch. 1, Fig. 1.8).

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Prop. 8.1, p. 153); in such games the SPE are exactly the backward-induction profiles (Lecture 3, slide 35). An SPE is written as complete strategies: player 2's strategy names a reply at *both* of her nodes, including the one off the path (Lecture 3, slide 25; Tadelis Def. 7.4, p. 138; Gibbons pp. 125–126 on outcome vs. equilibrium). All Nash equilibria of the normal form are found with the best-reply and indifference conditions (principle of indifference, Lecture 3, slides 10–12).

(c) Solution. Notation. C = Confess, N = Not confess. Payoffs (u_1, u_2) from Fig. 4.9: $(C,C) \mapsto (-5,-5)$, $(C,N) \mapsto (0,-15)$, $(N,C) \mapsto (-15,0)$, $(N,N) \mapsto (-1,-1)$. Player 2's strategy is a pair $s_2 = s_2(C)\, s_2(N)$: her reply after C, then after N. So $S_2 = \{CC, CN, NC, NN\}$ and $S_1 = \{C, N\}$.

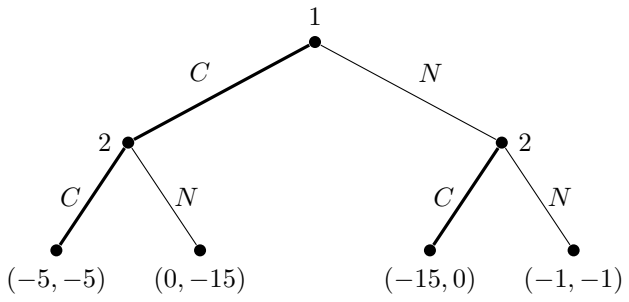


Fig. 4.9 redrawn; thick branches = backward-induction choices.

(a) Player 2's best replies.

$$\begin{aligned} \text{after } C: &\quad u_2(C,C) = -5 > -15 = u_2(C,N) \;\Rightarrow\; C; \\ \text{after } N: &\quad u_2(N,C) = 0 > -1 = u_2(N,N) \;\Rightarrow\; C. \end{aligned}$$

So $s_2^* = CC$: confess whatever player 1 did.

(b) Player 1's move and the SPE. Replacing each of player 2's nodes by its outcome,

$$u_1(C, s_2^*) = u_1(C,C) = -5 \;>\; -15 = u_1(N,C) = u_1(N, s_2^*),$$

so player 1 confesses. The SPE is

$$(s_1^*, s_2^*) = (C,\ (C \text{ after } C,\ C \text{ after } N)) = (C, CC),$$

with outcome (C,C) and payoffs $(-5,-5)$. It is unique: every comparison above is strict.

Normal form and all Nash equilibria (deepening).

	CC	CN	NC	NN
C	$\underline{-5},\ \underline{-5}$	$-5,\ \underline{-5}$	$\underline{0},\ -15$	$\underline{0},\ -15$
N	$-15,\ \underline{0}$	$\underline{-1},\ -1$	$-15,\ \underline{0}$	$-1,\ -1$

Underlined payoffs are best replies. *Pure:* only (C, CC) has both payoffs underlined. *Mixed:* let $p = \Pr_1(C)$.

- If $0 < p < 1$, player 2's payoffs are $-5p$ (CC), $-5p-(1-p)$ (CN), $-15p$ (NC), $-15p-(1-p)$ (NN); CC is the unique best reply, and against CC player 1 strictly prefers C $(-5 > -15)$, so $p = 1$. Contradiction.
- If $p = 0$, player 2's best replies are CC and NC (both reply C to N), and against either one C pays player 1 -5 or 0, more than -15. Contradiction.
- If $p = 1$, player 2's best replies are CC and CN (both reply C to C; they differ only at the unreached node). Let $w = \Pr_2(CN)$. Player 1 keeps C iff

$$u_1(C) = -5 \;\geq\; u_1(N) = -15(1-w) + (-1)w = -15 + 14w \iff w \leq \tfrac{5}{7}.$$

So the Nash equilibria are $\{(C,\ (1-w)\,CC + w\,CN) : 0 \leq w \leq \frac{5}{7}\}$, all with outcome (C,C). Only $w = 0$ is subgame perfect: for $w > 0$ player 2 plans N after N with positive probability, which is not a best reply in that subgame $(-1 < 0)$. In behavioral terms: player 2 plays C after C and, after N, plays N with probability at most $\frac{5}{7}$.

(c) Comparison. In the simultaneous game (Fig. 1.8) C is strictly dominant for each prisoner $(-5 > -15$ and $0 > -1)$, so the unique Nash equilibrium is (C,C) with $(-5,-5)$. The sequential version has the same outcome. Seeing player 1's move does not change player 2's choice, because C is her best reply to each of player 1's moves; player 1 therefore gains nothing from moving first.

(d) Interpretation. Observing the other's move matters only when the best reply depends on it. Here it does not: confessing is dominant at both of player 2's nodes, so the sequential game unravels exactly as the simultaneous one, and the Pareto-better (N, N) is still out of reach. The mixed Nash equilibria show what subgame perfection removes even here: a "promise" by player 2 to reciprocate silence (N after N) is cheap talk once player 1 has stayed silent, the Lecture 4 lesson that talk is cheap unless something makes it credible (Lecture 4, slide 11). Moving first is a commitment device only when it changes what the follower wants to do (Stackelberg, Lecture 3, slide 37).

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique result: path C, C with payoffs $(-5, -5)$), finds the pure Nash equilibria of the 2×4 normal form (only (C, CC)) and of the simultaneous game (only (C, C)), confirms on a grid of w in steps of $\frac{1}{70}$ that $(C, (1-w)CC + wCN)$ is a Nash equilibrium exactly when $w \leq \frac{5}{7}$, and that no Nash equilibrium puts weight on N. Output line: `munoz-garcia-4-3: SPE (C, CC); NE = (C, CC with prob >= 2/7)`. *Comparison with Muñoz-Garcia's answer* (pp. 160–162): it agrees on all three parts, SPNE $= \{(\text{Confess}, \text{Confess}, \text{Confess})\}$. One wording slip there: the third component is described as player 2's response "to player 1 selecting Confess (in Subgame 3)"; Subgame 3 follows Not Confess, so it is the reply to Not Confess. The book does not list the mixed Nash equilibria.

(f) Common mistakes.

- Writing the SPE as the outcome (C, C): player 2's strategy must say what she does after N too, so the SPE is (C, CC).
- Concluding that the sequential game has a different equilibrium because "player 2 can punish". Punishment after N is not a best reply, so it is not part of any SPE.
- Claiming that (C, CC) is the only Nash equilibrium: it is the only *pure* one, but player 2 may mix off the path ($w \leq \frac{5}{7}$).

Answer. (a) Player 2 confesses after Confess ($-5 > -15$) and after Not confess ($0 > -1$). (b) Player 1 confesses ($-5 > -15$); SPE $= (C, (C, C))$, outcome (C, C), payoffs $(-5, -5)$. Nash equilibria: $(C,\ (1-w)CC + wCN)$, $0 \leq w \leq \frac{5}{7}$; only $w = 0$ is subgame perfect. (c) No difference: the simultaneous game's unique equilibrium is also (C, C), since Confess is strictly dominant at every node.

Item 4: Tadelis Exercise 7.2

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

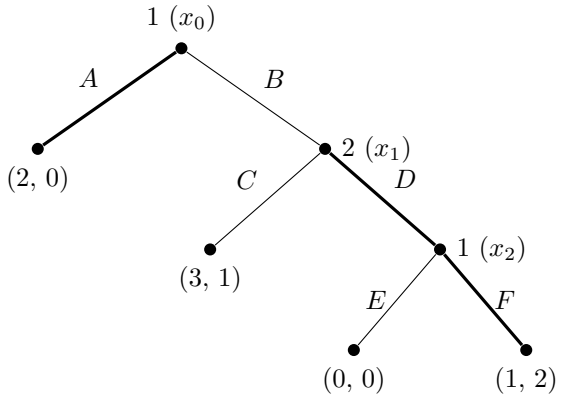
Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.2 "Strategies and Equilibrium", p. 147 [PDF 166]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Strategies and Equilibrium:** Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can then choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 can then choose E or F, and the game ends after each of these choices.
>
> a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?
>
> b. How many terminal nodes does the game have? How many information sets?
>
> c. How many pure strategies does each player have?
>
> d. Imagine that the payoffs following choice A by player 1 are $(2, 0)$, those following C by player 2 are $(3, 1)$, those following E by player 1 are $(0, 0)$, and those following F by player 1 are $(1, 2)$. What are the Nash equilibria of this game? Does one strike you as more "appealing" than the other? If so, explain why.

(a) What is asked. For the three-move game A/B, then C/D, then E/F: (a) its tree and type of information; (b) terminal nodes and information sets; (c) pure strategies per player; (d) with the given payoffs, all Nash equilibria and which one is more appealing.

(b) Method. Game tree and information sets (Lecture 3, slides 20–23; Tadelis §7.1, Defs. 7.1–7.2, pp. 133–135); perfect information (slide 24; Def. 7.3, p. 136); pure strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form of an extensive-form game (Tadelis §7.2.3, pp. 142–145); Nash equilibria by best-reply marking and indifference; the "appealing" one by sequential rationality and backward induction (slides 29–30; Tadelis §8.1, pp. 152–153).

(c) Solution. **(a) Tree.** Payoffs (u_1, u_2) from part (d) are written at the leaves.



Thick branches = backward-induction choices (part (d)).

Each decision node is alone in its information set (the mover always knows the history), so it is a game of **perfect information**.

(b) Counts. Terminal nodes: the leaves after A, C, E, F, so **4**. Information sets: $\{x_0\}$, $\{x_2\}$ for player 1 and $\{x_1\}$ for player 2, so **3** (two for player 1, one for player 2).

(c) Pure strategies. Player 1 picks an action at x_0 and at x_2, even when his first action (A) makes x_2 unreachable: $S_1 = \{AE, AF, BE, BF\}$, **4** strategies (2×2, item "Tadelis 7.1"). Player 2 has one information set: $S_2 = \{C, D\}$, **2** strategies.

(d) Nash equilibria. Normal form (rows player 1, columns player 2):

	C	D
AE	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
AF	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
BE	$\underline{3}$, $\underline{1}$	0, 0
BF	$\underline{3}$, 1	1, $\underline{2}$

Pure Nash equilibria (both payoffs underlined): (AE, D), (AF, D) and (BE, C). (BF, C) fails because against BF player 2 prefers D ($2 > 1$).

All Nash equilibria. Let $q = \Pr_2(C)$. Player 1's payoffs are

$$u_1(AE) = u_1(AF) = 2, \qquad u_1(BE) = 3q, \qquad u_1(BF) = 3q + (1 - q) = 1 + 2q.$$

- $q < \frac{1}{2}$: $2 > 1 + 2q > 3q$, so player 1 plays A (any mix of AE, AF); player 2's node is not reached, she is indifferent, and any such q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: AE, AF, BF tie at 2. If BF had positive weight, player 2's node would be reached only after BF, where D is strictly better ($2 > 1$), forcing $q = 0$. So player 1 stays on $\{AE, AF\}$. *Equilibria.*
- $\frac{1}{2} < q < 1$: $1 + 2q > \max\{2, 3q\}$, so BF is the unique best reply, and then $q = 0$. Contradiction.
- $q = 1$: BE and BF tie at $3 > 2$. With $w = \Pr_1(BF)$ (the rest on BE), player 2 gets 1 from C and $2w$ from D, so C is a best reply iff $w \leq \frac{1}{2}$. *Equilibria.*

Hence the set of Nash equilibria is

$$\{(\text{any mix of } AE, AF;\ q) : 0 \leq q \leq \tfrac{1}{2}\} \ \cup\ \{((1 - w)\, BE + w\, BF;\ C) : 0 \leq w \leq \tfrac{1}{2}\},$$

with outcome A and payoffs $(2, 0)$ in the first family and outcome (B, C), payoffs $(3, 1)$, in the second.

The appealing one. Backward induction: at x_2 player 1 picks F ($1 > 0$); at x_1 player 2, anticipating F, compares $C \mapsto 1$ with $D \mapsto 2$ and picks D; at x_0 player 1 compares $A \mapsto 2$ with $B \mapsto 1$ and picks A. The backward-induction profile, and the unique SPE, is (AF, D), outcome A, payoffs $(2, 0)$. It is the appealing one because every player's plan is optimal at every node, reached or not. The other two pure equilibria rely on player 1's plan to play E at x_2, which gives him $0 < 1$:

- in (BE, C) the plan E is a *threat* that scares player 2 into C; if x_2 were reached, player 1 would not carry it out;
- in (AE, D) the plan E is never tested and does not change the outcome, but it is still not sequentially rational.

(d) Interpretation. Player 1 would like to commit to E ("if you play D I will burn both our payoffs"), because that threat earns him 3 instead of 2. He cannot: at x_2 the damage is in the past and F is better. Nash equilibrium accepts the threat because x_2 is off the path; sequential rationality rejects it. This is the Brexit pattern of Lecture 4 (slides 8–13): several Nash equilibria are held up by threats nobody would execute, and only the backward-induction profile survives.

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique solution, outcome A, payoffs $(2, 0)$; continuation D then F after B), finds the three pure Nash equilibria of the 4×2 matrix, scans q on a grid of step $\frac{1}{120}$ plus $\frac{1}{2}, \frac{2}{3}$ (Nash equilibria exist exactly for $q \leq \frac{1}{2}$ and $q = 1$), and tests the $w \leq \frac{1}{2}$ condition on a grid. Output line: `tadelis-7-2: SPE (AF, D); pure NE (AE,D), (AF,D), (BE,C)`. Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 only three strategies (A, BE, BF). That is the reduced normal form; as Tadelis defines strategies, A must be completed with a plan at x_2.
- Calling (BE, C) the "best" equilibrium because it gives player 1 more. A Nash equilibrium is appealing when its threats are credible, not when it pays one player more.
- Counting 3 information sets as "3 decision nodes per player" or counting terminal nodes as information sets.

Answer. (a) Tree above; perfect information. (b) 4 terminal nodes; 3 information sets (2 for player 1, 1 for player 2). (c) Player 1: 4 (AE, AF, BE, BF); player 2: 2 (C, D). (d) Pure Nash equilibria (AE, D), (AF, D), (BE, C) (plus mixed ones: player 1 on A with $\Pr(C) \leq \frac{1}{2}$, or $(1 - w)BE + wBF$ with $w \leq \frac{1}{2}$ against C). The appealing one is (AF, D), the unique SPE: the others rest on the non-credible plan E.

Item 5: Muñoz-Garcia Ch. 4, Exercise 4

Foundations · difficulty 2/5

Statement (verbatim from the list, item 5)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 4 "Sequential Version of the Battle of the Sexes Game", p. 163 [PDF 175]. Reproduced verbatim (statement only; the book's worked answer follows it). *Note (not in the book):* part (a) as printed swaps the roles; the wife moves first, so read it as the husband's best responses to each of the wife's actions, which is what the book's answer computes.

> Consider the game tree in Fig. 4.13, representing the sequential-move version of the Battle of the Sexes game, where wife acts first, and observing its move husband responds.

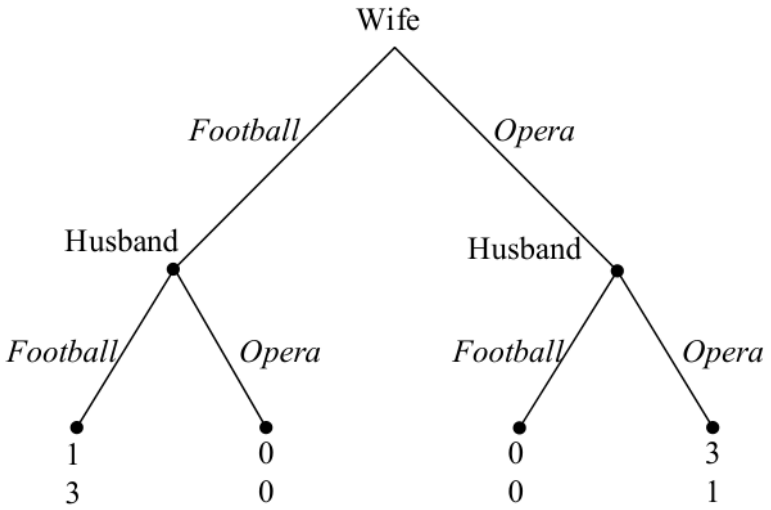


Fig. 4.13 Battle of the sexes sequential-move game

(a) Find the best responses of the wife, for each of the husband's actions.

(b) Find the husband's equilibrium action and describe the subgame perfect Nash equilibrium of the game.

(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

(a) What is asked. In the sequential battle of the sexes of Fig. 4.13 (the wife moves first, the husband observes her move): (a) the best replies of the second mover to each action of the first mover; (b) the equilibrium action of the first mover and the subgame-perfect equilibrium; (c) whether play differs from the simultaneous version.

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slides 59–64; slide 64 solves the same game with Alex and Chris; Tadelis §8.1, p. 153). Part (a) as printed asks for the wife's best replies, but the wife moves first; read it as the husband's best replies to each of the wife's actions, which is what the book's answer computes. A subgame-perfect equilibrium lists a husband's reply at both of his nodes (Lecture 3, slides 65–68).

(c) Solution. **Notation.** F = Football, O = Opera. Payoffs (wife, husband) from Fig. 4.13: $(F,F) \mapsto (1,3)$, $(F,O) \mapsto (0,0)$, $(O,F) \mapsto (0,0)$, $(O,O) \mapsto (3,1)$. The husband's strategy is a pair $s_H = s_H(F)\, s_H(O)$ (his reply after F, then after O): $S_H = \{FF, FO, OF, OO\}$; $S_W = \{F, O\}$.

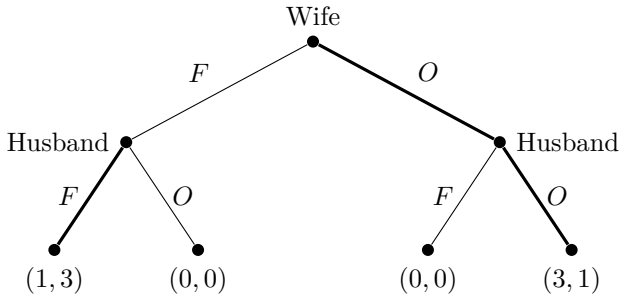


Fig. 4.13 redrawn; thick branches = backward-induction choices.

(a) The husband's best replies.

$$\begin{aligned} &\text{after } F: \quad u_H(F,F) = 3 > 0 = u_H(F,O) \;\Rightarrow\; F; \\ &\text{after } O: \quad u_H(O,O) = 1 > 0 = u_H(O,F) \;\Rightarrow\; O. \end{aligned}$$

The husband copies the wife: $s_H^* = FO$.

(b) The wife's action and the SPE. Replacing each husband node by his reply,

$$u_W(F, s_H^*) = u_W(F,F) = 1 \;<\; 3 = u_W(O,O) = u_W(O, s_H^*),$$

so the wife chooses O. The subgame-perfect equilibrium is

$$(s_W^*, s_H^*) = (O,\ (F \text{ after } F,\ O \text{ after } O)) = (O, FO),$$

with outcome (O,O) and payoffs $(3,1)$. It is unique: both comparisons are strict.

Normal form and Nash equilibria that are not subgame perfect.

	FF	FO	OF	OO
F	$\underline{1},\ \underline{3}$	$1,\ \underline{3}$	$\underline{0},\ 0$	$0,\ 0$
O	$0,\ 0$	$\underline{3},\ \underline{1}$	$\underline{0},\ 0$	$\underline{3},\ \underline{1}$

Underlined payoffs are best replies. The pure Nash equilibria are (F, FF), (O, FO) and (O, OO). Only (O, FO) is subgame perfect. (F, FF) rests on the husband's threat to go to Football even after the wife chooses Opera (payoff 0 for him, against 1 from Opera), and (O, OO) on his threat to go to Opera even after Football (payoff 0 against 3): neither threat is a best reply in its subgame, so neither is credible.

(c) Comparison. The simultaneous game has two pure equilibria, (F,F) with $(1,3)$ and (O,O) with $(3,1)$, and a mixed one (Lecture 3, slides 27–34); which is played is undetermined. In the sequential game the husband observes the wife and copies her, so the wife can pick her favorite: the unique SPE outcome is (O,O), her best equilibrium of the simultaneous game. Moving first is an advantage here: the wife earns 3 instead of 1 or the mixed-equilibrium payoff. The Football outcome (F,F) is a Nash equilibrium of the sequential game too, but only through a non-credible threat.

(d) Interpretation. The first-mover advantage comes from commitment: once the wife has moved, the husband can only respond, and responding means coordinating on her choice. If the husband moved first the opposite outcome would arise. The advantage depends on the *coordination* structure: in the prisoner's dilemma (item 3) moving first was worth nothing.

(e) Verification. `check_L03.py` runs backward induction on the tree (the unique solution is path O, O with payoffs $(3, 1)$), enumerates the pure Nash equilibria of the 2×4 normal form and of the simultaneous game, and checks that exactly one of the three is a backward-induction profile. Output line: `munoz-garcia-4-4: SPE (O, FO); pure NE (F,FF), (O,FO), (O,OO).` *Comparison with Muñoz-Garcia's answer* (pp. 163–165): it agrees on all three parts, giving SPNE = {Opera, (Football, Opera)} with the husband's reply in Subgame 2 (after Football) first and Subgame 3 (after Opera) second, and it states that the husband moving first would reverse the outcome. The book does not list the non-subgame-perfect Nash equilibria.

(f) Common mistakes.

- Answering (a) as printed (the wife's best reply to the husband's action): the husband observes the wife, not the reverse.
- Writing the SPE as the outcome (O, O) or as "wife Opera, husband Opera": the husband's strategy must also say what he does after Football, so the SPE is (O, FO).
- Concluding that (F, F) cannot be an equilibrium of the sequential game: it is a Nash equilibrium (sustained by FF), just not a subgame-perfect one.

Answer. (a) The husband replies Football after Football ($3 > 0$) and Opera after Opera ($1 > 0$). (b) The wife chooses Opera ($3 > 1$); SPE $= (O, (F, O))$, outcome (O, O), payoffs $(3, 1)$. Nash equilibria: (F, FF), (O, FO), (O, OO); only (O, FO) is subgame perfect. (c) Yes: the simultaneous game has two pure equilibria and a mixed one; the sequential game's unique SPE outcome is the wife's preferred one, (O, O) (first-mover advantage).

Item 6: Tadelis Exercise 7.6

Foundations · difficulty 1/5

Statement (verbatim from the list, item 6)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.6 "Entering an Industry", pp. 148–149 [PDF 167–168]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Entering an Industry:** A firm (player 1) is considering entering an established industry with one incumbent firm (player 2). Player 1 must choose whether or not to enter the industry. If player 1 enters the industry then player 2 can either accommodate the entry or fight the entry by waging a price war. Player 1's most-preferred outcome is entering with player 2 not fighting, and its least-preferred outcome is entering with player 2 fighting. Player 2's most-preferred outcome is player 1 not entering, and its least-preferred outcome is player 1 entering with player 2 fighting.
>
> a. Model this as an extensive-form game tree (choose payoffs that represent the preferences).
>
> b. How many pure strategies does each player have?
>
> c. Find all the Nash equilibria of this game.

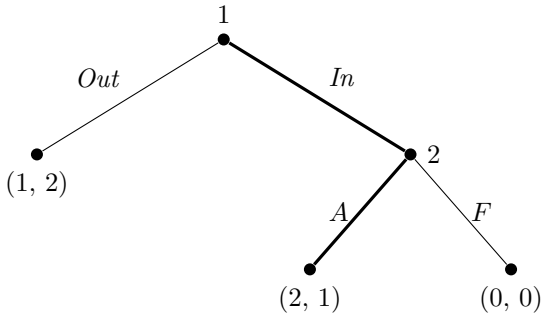
(a) What is asked. (a) A game tree with payoffs that represent the stated rankings; (b) the number of pure strategies of each player; (c) all Nash equilibria.

(b) Method. Extensive form and game trees (Tadelis §7.1, pp. 130–137); pure strategies as complete plans (§7.2.1, p. 137); the normal-form representation of a game tree (§7.2.3, p. 143); Nash equilibrium and paths of play, on and off the equilibrium path (§7.3 and Def. 7.8, pp. 145–146). The question stops at Nash equilibrium; which equilibrium survives sequential rationality is Tadelis §8.1 (Def. 8.1, p. 152) and Lecture 4, slides 13 and 17 (the entry-deterrence logic of the Brexit case, slides 9–11).

(c) Solution. **(a) Tree.** Ordinal payoffs suffice, since only the rankings are given. Write *In* and *Out* for player 1's choice and A (accommodate), F (fight) for player 2's. The rankings are

$$\text{player 1: } (In, A) \succ Out \succ (In, F), \qquad \text{player 2: } Out \succ (In, A) \succ (In, F).$$

Player 1's middle outcome must be *Out* and player 2's must be (In, A), the only outcomes not named as best or worst. Payoffs $2, 1, 0$ in that order give



Payoffs (u_1, u_2); thick branches = backward-induction choices (used below).

Any numbers with the same rankings give the same answers to (b) and (c).

(b) Pure strategies. Each player has one information set with two actions, so $S_1 = \{In, Out\}$ and $S_2 = \{A, F\}$: two pure strategies each. Player 2's strategy is a plan for the node after entry, and it is part of the profile even when player 1 stays out.

(c) Nash equilibria. The normal form, with best replies underlined:

	A	F
In	$\underline{2},\ \underline{1}$	$0,\ 0$
Out	$1,\ \underline{2}$	$\underline{1},\ \underline{2}$

Pure. (In, A) and (Out, F). Against A, In gives $2 > 1$; against F, Out gives $1 > 0$. Against In, A gives $1 > 0$; against Out, player 2 gets 2 with either action, so F is a (weak) best reply.

Mixed. Let $p = \Pr(In)$ and $f = \Pr(F)$.

- If $p > 0$, then $u_2(A) - u_2(F) = p \cdot (1 - 0) > 0$, so $f = 0$, and then In is player 1's unique best reply ($2 > 1$): $p = 1$. This is (In, A).
- If $p = 0$, player 2 is indifferent, so any f is a best reply. Out is a best reply for player 1 iff

$$u_1(Out) = 1 \ \geq \ u_1(In) = 2(1 - f) + 0 \cdot f \iff f \geq \tfrac{1}{2}.$$

So the set of all Nash equilibria is

$$\{(In, A)\} \ \cup \ \{(Out,\ \sigma_2) : \sigma_2(F) \geq \tfrac{1}{2}\},$$

a single entry equilibrium plus a continuum of deterrence equilibria, (Out, F) among them.

Which are sequentially rational. At player 2's node, A gives $1 > 0$. Every equilibrium of the second family puts weight $f \geq \frac{1}{2}$ on F at a node that is off the equilibrium path (Def. 7.8), where F is not a best response. Backward induction (thick branches) keeps only (In, A), which is therefore the unique subgame-perfect equilibrium (Tadelis Fact, p. 158; slide 17).

(d) Interpretation. "Fight if you enter" deters entry only as long as nobody tests it. Once the entrant is in, the price war hurts the incumbent ($0 < 1$), so the threat is not credible, and an entrant who reasons backward enters. Nash equilibrium cannot see this: in (Out, F) the war is never fought, so planning it costs player 2 nothing. To deter entry, the incumbent needs a commitment made before the entry decision that makes fighting its best response after entry (excess capacity, a sunk investment, a contract). The Brexit case uses the same logic: the UK's sunk preparation makes rejecting a bad deal credible (slides 9–11).

(e) Verification. `check_D.py` builds the tree, enumerates the 2×2 pure profiles, tests every one for Nash and for subgame perfection by definition, and tests every mixed profile on a $\frac{1}{100}$ grid in both probabilities with exact fractions. Output: "Tadelis 7.6: 2 strategies each; pure NE (In,A), (Out,F); all NE = (In,A) and Out with Pr(F) >= 1/2; SPE (In,A) only". Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 the payoff ordering of a "most-preferred / least-preferred" list without placing Out in the middle. The middle outcome is implied, not stated.
- Answering (c) with (In, A) only. The question asks for Nash equilibria, and (Out, F) is one, as are all (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$.

- Calling (Out, F) not Nash because "player 2 would not fight". Player 2 is indifferent on the path; the flaw is off the path, which is what subgame perfection adds.

Answer. (a) Tree above, with $u(Out) = (1, 2)$, $u(In, A) = (2, 1)$, $u(In, F) = (0, 0)$. (b) Two pure strategies each: $\{In, Out\}$ and $\{A, F\}$. (c) Pure Nash equilibria (In, A) and (Out, F); with mixing, all Nash equilibria are (In, A) and (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$. Only (In, A) is subgame perfect: the threat to fight is not credible.

Item 7: Muñoz-Garcia Ch. 4, Exercise 9

Foundations · difficulty 2/5

Statement (verbatim from the list, item 7)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 9 "Centipede Game", pp. 176–177 [PDF 188–189]. Reproduced verbatim (statement only; the book's worked answer follows it). The figure is printed with the book's subgame boxes already drawn. *Note (not in the book):* the text says depositor 1 moves at the 100th stage, but Fig. 4.31 and the book's answer have depositor 2 as the last mover, choosing between $(99, 101)$ and $(100, 100)$; follow the figure.

> Consider the game tree in Fig. 4.31, representing the so-called "centipede game" between depositor 1 and 2. In the first stage, depositor 1 chooses whether to take the money in the saving account, which yields a payoff of \$2 for him and \$0 for the other depositor, or keep the money in the account. If depositor 1 keeps the money in the account, the game continues to stage 2 where depositor 2 can take the money or keep it in the account for one more period. If he takes the money, he receives a payoff of \$3 while depositor 1 receives only \$1. If depositor 2 keeps the money in the account, then it's depositor 1's turn to choose whether to take the money or keep it for one more period. The game continues with depositors alternatively choosing whether to take the money until the 100th stage, when it is depositor 1's turn to move. Since the game tree has 100 branches pointing down, it is often called the "centipede" game since the game tree looks like an insect with 100 legs.
>
> a. Find the best response of the last mover.
>
> b. Anticipating equilibrium behavior in the 100th stage, find the best response of the previous-to-last mover (the depositor acting in stage 99).

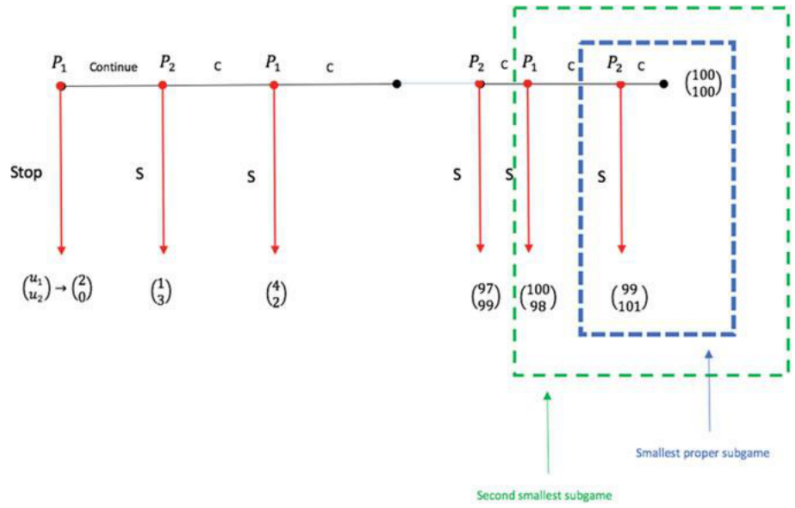


Fig. 4.31 Centipede sequential-form game

> c. Using your results in parts (a) and (b), find the subgame perfect Nash equilibrium of the game.

(a) What is asked. In the 100-stage centipede of Fig. 4.31: (a) the last mover's best reply; (b) the best reply of the previous-to-last mover given (a); (c) the subgame-perfect equilibrium.

(b) Method. Backward induction on a long chain (Lecture 3, slides 59–64; Tadelis §8.3.1 on the centipede, pp. 159–160): the last mover's choice is the same at every history, so the argument for stage t is the argument for stage $t+1$ shifted by one. The text of the exercise says depositor 1 moves at the 100th stage, but Fig. 4.31 and the book's answer have depositor 2 as the last mover, choosing between $(99, 101)$ and $(100, 100)$; this solution follows the figure.

(c) Solution. The payoffs. Read off Fig. 4.31, with $t = 1, \ldots, 100$ the stage. Depositor 1 moves at odd stages, depositor 2 at even stages. If the mover at stage t stops (*Stop*), the payoffs (u_1, u_2) are

$$t = 2k-1 \text{ (depositor 1)}: \ (2k,\ 2k-2), \qquad t = 2k \text{ (depositor 2)}: \ (2k-1,\ 2k+1),$$

for $k = 1, \ldots, 50$: $(2,0)$, $(1,3)$, $(4,2)$, …, $(97,99)$ at $t = 98$, $(100,98)$ at $t = 99$, $(99,101)$ at $t = 100$. If depositor 2 continues at stage 100, the game ends at $(100, 100)$.

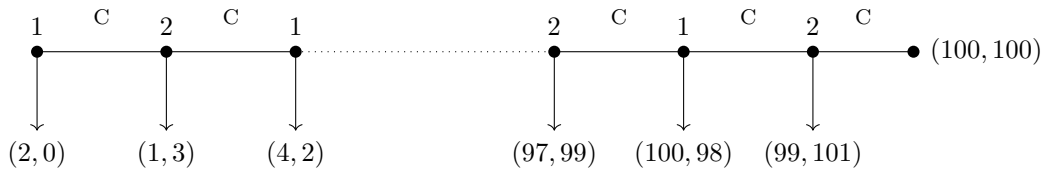


Fig. 4.31 redrawn; down arrows are *Stop*, horizontal arcs are *Continue* (stages 1–3 and 98–100 shown).

(a) The last mover (stage 100, depositor 2). The subgame is a single choice:

$$u_2(\text{Stop}) = 101 > 100 = u_2(\text{Continue}) \ \Rightarrow \ \text{Stop}.$$

This is the best reply whatever happened before: the last node is a proper subgame, so the history does not enter the comparison.

(b) Stage 99 (depositor 1). Anticipating Stop at stage 100,

$$u_1(\text{Stop}) = 100 > 99 = u_1(\text{Continue}) \ \Rightarrow \ \text{Stop},$$

because continuing leads to $(99, 101)$, in which depositor 1 gets 99.

(c) Induction to the root. Suppose that every mover from stage $t+1$ on stops. At stage t:

- if depositor 1 moves ($t = 2k-1$), stopping pays $2k$, and continuing pays what he receives when depositor 2 stops at stage $2k$, namely $2k-1$: $2k > 2k-1$;
- if depositor 2 moves ($t = 2k$), stopping pays $2k+1$, and continuing pays what he receives when depositor 1 stops at stage $2k+1$, namely $2k$ (the payoff $(2k+2, 2k)$): $2k+1 > 2k$ (for $t = 100$, $k = 50$, continuing pays $100 = 2k$ in the same way).

The comparison always favors Stop by exactly one dollar, so Stop is optimal at every stage, down to the first: $u_1(\text{Stop}) = 2 > 1 = u_1(\text{Continue})$ (continuing would let depositor 2 stop at $(1,3)$). The unique SPE is the strategy profile in which *each depositor plays Stop at every one of his nodes*, on and off the equilibrium path; the outcome is $(2,0)$ at stage 1. It is the same for any length of centipede. Uniqueness holds because every comparison is strict.

(d) Interpretation. Each player would gain from the other's cooperation: $(100, 100)$ gives depositor 1 fifty times his equilibrium payoff and depositor 2 infinitely more. But each stop is a best reply only because the other is expected to stop one step later, and the chain of expectations starts from a last stage where cooperation has no future. Nash equilibrium alone would not generate the paradox with such sharpness; backward induction does. The unraveling is the same logic as the finitely repeated prisoner's dilemma (Lecture 5 and Tadelis Ch. 8–10).

(e) Verification. `check_L03.py` builds the 100-stage tree with the payoff pattern above (checking the printed payoffs $(2,0)$, $(1,3)$, $(4,2)$, $(97,99)$, $(100,98)$, $(99,101)$, $(100,100)$), runs backward induction with exact integers, and asserts that every mover stops at every node, that each stop beats continuing by exactly one dollar, and that the outcome is $(2,0)$; it repeats the check for lengths $2,4,\ldots,20$. Output line: `munoz-garcia-4-9: SPE = Stop at every stage; outcome (2,0); margin 1 everywhere`. *Comparison with Muñoz-Garcia's answer* (pp. 177): it agrees, ending with SPNE $=(\text{Stop}_t,\text{Stop}_t)$ in every period t.

(f) Common mistakes.

- Following the text rather than the figure on who moves last: the figure has no payoffs for a depositor 1 move at stage 100.
- Describing the SPE by its outcome ("depositor 1 stops at once"): the strategy also says that every later mover would stop if reached.
- Arguing that a patient depositor 2 "should wait": at stage 2 he compares 3 now with 2 later, not with 100.

Answer. (a) At stage 100 depositor 2 stops ($101 > 100$). (b) At stage 99 depositor 1 stops ($100 > 99$). (c) The unique SPE has every depositor stopping at every node; the outcome is depositor 1 stopping at stage 1, payoffs $(2,0)$.

Item 8: Tadelis Exercise 7.4

Exam · difficulty 3/5

Statement (verbatim from the list, item 8)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.4 "Centipedes", p. 148 [PDF 167]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Centipedes:** Imagine a two-player game that proceeds as follows. A pot of money is created with \$6 in it initially. Player 1 moves first, then player 2, then player 1 again, and finally player 2 again. At each player's turn to move, he has two possible actions: grab (G) or share (S). If he grabs he gets $\frac{2}{3}$ of the current pot of money, the other player gets $\frac{1}{3}$ of the pot, and the game ends. If he shares then the size of the current pot is multiplied by $\frac{3}{2}$ and the next player gets to move. At the last stage at which player 2 moves, if he chooses S then the pot is still multiplied by $\frac{3}{2}$, player 2 gets $\frac{1}{3}$ of the pot, and player 1 gets $\frac{2}{3}$ of the pot.
>
> a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?
> b. How many terminal nodes does the game have? How many information sets?
> c. How many pure strategies does each player have?
> d. Find the Nash equilibria of this game. How many outcomes can be supported in equilibrium?
> e. Now imagine that at the last stage at which player 2 moves, if he chooses to share then the pot is equally split among the players. Does your answer to part (d) change?

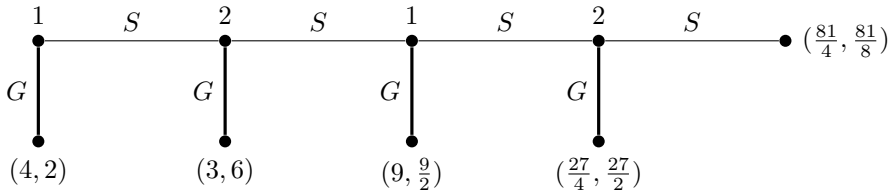
(a) What is asked. In the four-move grab/share game with a pot of \$6 growing by $\frac{3}{2}$: (a) tree and type of information; (b) terminal nodes and information sets; (c) pure strategies; (d) Nash equilibria and how many outcomes they support; (e) the same when sharing at the last node splits the pot equally.

(b) Method. Tree, information sets and perfect information (Lecture 3, slides 20–24; Tadelis §7.1, pp. 133–136); strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form and its Nash equilibria (Tadelis §7.2.3, pp. 142–145); backward induction to identify the sequentially rational one (slides 29–30; Tadelis §8.1, pp. 152–153; the centipede game itself is Tadelis §8.3.1, pp. 159–160). Nash equilibrium constrains play only on the path, so the outcome argument proceeds forward from the last node that is reached with positive probability.

(c) Solution. Pot and payoffs. The pot before move $k = 1, 2, 3, 4$ is $6 \cdot (\frac{3}{2})^{k-1}$: 6, 9, $\frac{27}{2}$, $\frac{81}{4}$; after a share at move 4 it is $\frac{243}{8}$. A grab gives the mover $\frac{2}{3}$ of the current pot and the other player $\frac{1}{3}$. Payoffs (u_1, u_2):

Terminal history	pot	u_1	u_2
G (1 grabs at move 1)	6	4	2
SG (2 grabs at move 2)	9	3	6
SSG (1 grabs at move 3)	$\frac{27}{2}$	9	$\frac{9}{2}$
$SSSG$ (2 grabs at move 4)	$\frac{81}{4}$	$\frac{27}{4}$	$\frac{27}{2}$
$SSSS$ (2 shares at move 4)	$\frac{243}{8}$	$\frac{81}{4}$	$\frac{81}{8}$

(a) Tree. Perfect information: every information set is a single node.



Thick branches = backward-induction choices (part (d)).

(b) Counts. Terminal nodes: **5** (four grabs and the final share). Information sets: **4**, two per player (moves 1 and 3 for player 1, moves 2 and 4 for player 2).

(c) Pure strategies. Each player has 2 information sets with 2 actions: $2^2 = 4$ pure strategies each. Write $s_1 = (\text{move } 1, \text{move } 3)$ and $s_2 = (\text{move } 2, \text{move } 4)$: $S_1 = S_2 = \{GG, GS, SG, SS\}$.

(d) Nash equilibria. *Backward induction first.* Move 4: player 2 grabs ($\frac{27}{2} > \frac{81}{8}$). Move 3: player 1 grabs ($9 > \frac{27}{4}$). Move 2: player 2 grabs ($6 > \frac{9}{2}$). Move 1: player 1 grabs ($4 > 3$). SPE $= (GG, GG)$, payoffs $(4, 2)$.

Normal form (rows player 1, columns player 2):

	GG	GS	SG	SS
GG	$\underline{4}$, $\underline{2}$	$\underline{4}$, $\underline{2}$	4, $\underline{2}$	4, $\underline{2}$
GS	$\underline{4}$, $\underline{2}$	$\underline{4}$, $\underline{2}$	4, $\underline{2}$	4, $\underline{2}$
SG	3, $\underline{6}$	3, $\underline{6}$	$\underline{9}$, $\underline{\frac{9}{2}}$	9, $\underline{\frac{9}{2}}$
SS	3, 6	3, 6	$\underline{\frac{27}{4}}$, $\underline{\frac{27}{2}}$	$\underline{\frac{81}{4}}$, $\underline{\frac{81}{8}}$

(Underlined payoffs are best replies. Outside the top-left block no cell has both payoffs underlined: in row SS player 2's best reply is SG, against which player 1 prefers SG, against which player 2 prefers to grab at move 2.) The **pure Nash equilibria** are the four profiles $\{GG, GS\} \times \{GG, GS\}$: player 1 grabs at once and player 2 plans to grab at move 2; what either plans at the later, unreached node is free. Every one gives the outcome G with $(4, 2)$.

Mixed equilibria: still one outcome. Work forward along a Nash equilibrium, using that every node reached with positive probability must be played optimally given what follows. (i) If move 4 is reached, player 2 grabs there ($\frac{27}{2} > \frac{81}{8}$). (ii) So if move 3 is reached, sharing gives player 1 $\frac{27}{4}$ against 9 from grabbing: he grabs, and move 4 is never reached. (iii) So if move 2 is reached, sharing gives player 2 $\frac{9}{2}$ against 6 from grabbing: she grabs, and move 3 is never reached. (iv) So sharing at move 1 gives player 1 $3 < 4$: he grabs. Hence **every** Nash equilibrium has player 1 grabbing at move 1: **one outcome**, $(4, 2)$. The mixed equilibria differ only off the path. With $g = \Pr_2(G \text{ at move } 2)$ and $h = \Pr_2(G \text{ at move } 4)$, player 1's best deviation (share, then the better of grabbing for 9 or sharing for $M = \max\{9,\ \frac{27}{4}h + \frac{81}{4}(1-h)\}$) must not beat 4:

$$3g + M(1-g) \le 4 \iff g \ge 1 - \frac{1}{M-3},$$

e.g. $g \ge \frac{5}{6}$ if $h = 1$ and $g \ge \frac{65}{69}$ if $h = 0$.

(e) Equal split at the end. Sharing at move 4 now gives each player $\frac{1}{2} \cdot \frac{243}{8} = \frac{243}{16} \approx 15.19$. Backward induction: move 4: share ($\frac{243}{16} > \frac{27}{2} = \frac{216}{16}$); move 3: share ($\frac{243}{16} > 9$); move 2: share ($\frac{243}{16} > 6$); move 1: share ($\frac{243}{16} > 4$). SPE $= (SS, SS)$ with payoffs $(\frac{243}{16}, \frac{243}{16})$. The answer to (d) changes: the pure Nash equilibria are now (SS, SS) and, as before, the four profiles $\{GG, GS\} \times \{GG, GS\}$ (player 2's plan to grab at move 2 still deters player 1, and it is never tested). So **two outcomes** are supported: grab at once, $(4, 2)$, and share to the end, $(\frac{243}{16}, \frac{243}{16})$.

Grabbing at move 2, 3 or 4 is never an equilibrium outcome: the grabber at move 4 would rather share, the grabber at move 3 would be pre-empted at move 2 (player 2 compares 6 with $\frac{9}{2}$), and a grab at move 2 leaves player 1 with $3 < 4$.

(d) Interpretation. With the original split the mover at the last node always grabs, and the anticipation unravels cooperation back to the first move: the pot that could have grown to \$30 is grabbed at \$6. Changing only the *last* node's rule (an equal split) turns the last mover into someone who wants to continue, and the whole chain of reasoning reverses. Credibility at the end of the game drives behavior at the start, the backward-induction lesson of Lecture 3 and the reason why commitments that change the final payoffs (Lecture 4, slide 11) are so powerful.

(e) Verification. `check_B_L03.py` builds the tree with exact fractions (pots $6, 9, \frac{27}{2}, \frac{81}{4}, \frac{243}{8}$), runs backward induction (unique solution: grab, $(4, 2)$), builds the 4×4 normal form and finds exactly the four pure Nash equilibria, all with outcome $(4, 2)$; it checks the bound on g at $h \in \{0, \frac{1}{2}, 1\}$ on both sides ($\frac{5}{6}$, $\frac{65}{69}$); for (e) it finds the SPE all-share with $(\frac{243}{16}, \frac{243}{16})$ and the five pure Nash equilibria with two outcomes. Output line: `tadelis-7-4: (d) 4 pure NE, one outcome (4, 2); (e) SPE all-share 243/16 each; NE outcomes 2`. Tadelis prints no answer.

(f) Common mistakes.

- Answering "one Nash equilibrium" in (d): there are four pure ones (and a continuum of mixed ones); it is the *outcome* that is unique.
- Misreading the last-node rule in (d): if player 2 shares at move 4 she gets only $\frac{1}{3}$ of the pot, $\frac{81}{8} < \frac{27}{2}$, so she grabs.
- In (e), concluding that sharing is the only equilibrium outcome: the immediate grab is still a (non-subgame-perfect) Nash equilibrium.

Answer. (a) Tree above; perfect information. (b) 5 terminal nodes; 4 information sets (2 each). (c) 4 pure strategies each: GG, GS, SG, SS. (d) Pure Nash equilibria: $\{GG, GS\} \times \{GG, GS\}$; every Nash equilibrium (mixed ones too) has player 1 grabbing at once: one outcome, $(4, 2)$; SPE (GG, GG). (e) Yes: SPE (SS, SS) with $(\frac{243}{16}, \frac{243}{16})$; pure Nash equilibria (SS, SS) and $\{GG, GS\} \times \{GG, GS\}$; two outcomes.

Item 9: Tadelis Exercise 7.9

Exam · difficulty 4/5

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.9 "The Dean's Dilemma", pp. 149–150 [PDF 168–169]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

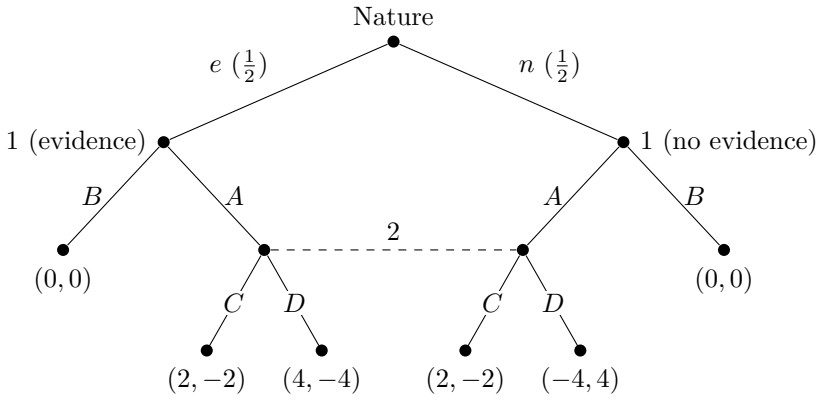
> **The Dean's Dilemma:** A student stole the DVD player from the student lounge. The dean of students (player 1) suspects the student (player 2) and begins collecting evidence. However, evidence collection is a random process, and concrete evidence will be available to the dean only with probability $\frac{1}{2}$. The student knows that the evidence-gathering process is under way but does not know whether the dean has collected evidence or not. The game proceeds as follows: The dean realizes whether or not he has evidence and can then choose his action, whether to accuse the student (A) or bounce the case (B) and forget it. Once accused, the student has two options: he can either confess (C) or deny (D). Payoffs are realized as follows: If the dean bounces the case then both players get 0. If the dean accuses the student and the student confesses, the dean gains 2 and the student loses 2. If the dean accuses the student and the student denies, then payoffs depend on the evidence: If the dean has no evidence then he loses face, which gives him a loss of 4, while the student gains glory, which gives him a payoff of 4. If, however, the dean has evidence then he is triumphant and gains 4, while the student is put on probation and loses 4.
>
> a. Draw the game tree that represents the extensive form of this game.
> b. Write down the matrix that represents the normal form of the extensive form you drew in (a).
> c. Solve for the Nash equilibria of the game.

(a) What is asked. The dean learns whether he has evidence (probability $\frac{1}{2}$), then accuses (A) or bounces (B); an accused student, who does not know whether there is evidence, confesses (C) or denies (D). (a) Game tree; (b) normal-form matrix; (c) Nash equilibria.

(b) Method. A move of Nature at the root and information sets for what a player does not observe (Lecture 3, slides 21–23; Tadelis §7.1, Def. 7.2, p. 135; Nature as in the Brexit tree, Lecture 4, slides 5–6). Strategies as plans over information sets (Lecture 3, slide 25), with expected payoffs over Nature's move in the normal form (Tadelis §7.2.3, pp. 142–145). Nash equilibria by dominance, best replies and the principle of indifference (Lecture 3, slides 10–12).

(c) Solution. Notation. Nature draws e (evidence) or n (no evidence), each with probability $\frac{1}{2}$. The dean (player 1) observes it. Payoffs (u_1, u_2) = (dean, student): $B \mapsto (0, 0)$; $A, C \mapsto (2, -2)$; A, D with evidence $\mapsto (4, -4)$; A, D without evidence $\mapsto (-4, 4)$. Note that $u_1 + u_2 = 0$ at every leaf: the game is zero-sum.

(a) Tree. The student's two nodes (after eA and after nA) form one information set: he knows he was accused, not whether there is evidence.



(b) Normal form. The dean has two information sets, so a strategy is a pair (action with evidence, action without): $S_1 = \{AA, AB, BA, BB\}$; $S_2 = \{C, D\}$. Each cell is $\frac{1}{2}\times$(payoff after e) $+\frac{1}{2}\times$(payoff after n). For example, $u(AB, D) = \frac{1}{2}(4, -4) + \frac{1}{2}(0, 0) = (2, -2)$ and $u(AA, D) = \frac{1}{2}(4, -4) + \frac{1}{2}(-4, 4) = (0, 0)$.

	C	D
AA	$\underline{2}$, -2	0, $\underline{0}$
AB	1, $\underline{-1}$	$\underline{2}$, -2
BA	1, -1	-2, $\underline{2}$
BB	0, $\underline{0}$	0, $\underline{0}$

(c) Nash equilibria.

Step 1: no pure equilibrium. No cell has both payoffs underlined: against C the dean plays AA, against which the student denies; against D the dean plays AB, against which the student confesses. The best replies cycle.

Step 2: dominated strategies. BA is strictly dominated by AA ($2 > 1$, $0 > -2$): accusing only when there is no evidence is never sensible. BB is strictly dominated by AB ($1 > 0$, $2 > 0$). So in any equilibrium the dean accuses whenever he has evidence and mixes only between AA and AB, i.e. over what he does *without* evidence.

Step 3: indifference. Let $q = \Pr_2(C)$ and $p = \Pr_1(AA)$ (with $1-p$ on AB). The dean is indifferent when

$$u_1(AA) = 2q \;=\; q + 2(1-q) = u_1(AB) \;\Rightarrow\; 2q = 2 - q \;\Rightarrow\; q = \tfrac{2}{3}, \tag{1}$$

and the student is indifferent when

$$u_2(C) = -2p - (1-p) \;=\; 0 \cdot p - 2(1-p) = u_2(D) \;\Rightarrow\; -1 - p = -2 + 2p \;\Rightarrow\; p = \tfrac{1}{3}. \tag{2}$$

At $q = \frac{2}{3}$ the dean's payoffs are $AA, AB : \frac{4}{3}$, $BA : 3q - 2 = 0$, $BB : 0$, so mixing over AA and AB is optimal. The unique Nash equilibrium is

$$\sigma_1^* = \tfrac{1}{3}\,AA + \tfrac{2}{3}\,AB, \qquad \sigma_2^* = \tfrac{2}{3}\,C + \tfrac{1}{3}\,D, \qquad (u_1, u_2) = (\tfrac{4}{3},\ -\tfrac{4}{3}).$$

Uniqueness. For $q < \frac{2}{3}$ only AB is a best reply and the student would then confess ($-1 > -2$), so $q = 1$; for $q > \frac{2}{3}$ only AA is a best reply and the student would deny ($0 > -2$), so $q = 0$; at $q = \frac{2}{3}$ only $p = \frac{1}{3}$ keeps the student indifferent.

Behavioral reading. With evidence the dean always accuses; without evidence he accuses (bluffs) with probability $\frac{1}{3}$. An accused student therefore faces evidence with probability $\frac{1/2}{1/2 + 1/2 \cdot 1/3} = \frac{3}{4}$, and at that belief C and D both give -2 (the check: $\frac{3}{4}(-4) + \frac{1}{4}(4) = -2$).

(d) Interpretation. The dean must bluff sometimes: if he accused only with evidence, every accusation would reveal evidence and the student would confess, which in turn makes accusing without evidence profitable. The equilibrium bluffing rate keeps the student exactly indifferent. Because the game is zero-sum, $\frac{4}{3}$ is its value: the dean cannot guarantee more, the student cannot hold him to less. The structure (Nature, a privately informed first mover, a responder who sees only the action) is the one Lecture 4 used for the Brexit negotiation, and it is the bridge to games of incomplete information later in the course.

(e) Verification. `check_B_L03.py` builds the matrix from the tree with exact fractions (it equals the one above), checks the zero-sum property and that there is no pure equilibrium, scans q on a grid of step $\frac{1}{300}$ (an equilibrium exists only at $q = \frac{2}{3}$), verifies $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$ with the exact Nash test, and solves (2) with sympy. Output line: `tadelis-7-9: unique NE dean 1/3 AA + 2/3 AB, student C w.p. 2/3; value 4/3`. Tadelis prints no answer.

(f) Common mistakes.

- Giving the dean only two strategies (A, B): he chooses after observing Nature, so a strategy is a pair of actions.
- Drawing the student's two nodes as separate information sets: then he would know whether there is evidence, and the game would change completely.
- Using payoffs conditional on one state in the matrix instead of the $\frac{1}{2}$–$\frac{1}{2}$ expectation.

Answer. (a) Tree above, with the student's nodes in one information set. (b) Matrix above, rows AA, AB, BA, BB (action with evidence, without). (c) No pure equilibrium; unique Nash equilibrium $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$: the dean always accuses with evidence and bluffs w.p. $\frac{1}{3}$ without it; the student confesses w.p. $\frac{2}{3}$; payoffs $(\frac{4}{3}, -\frac{4}{3})$.

Item 10: Tadelis Exercise 7.10

Exam · difficulty 4/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 7.10 "Perfect and Imperfect Recall", p. 150 [PDF 169]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

Perfect and Imperfect Recall: Consider the game depicted in Figure 7.14.

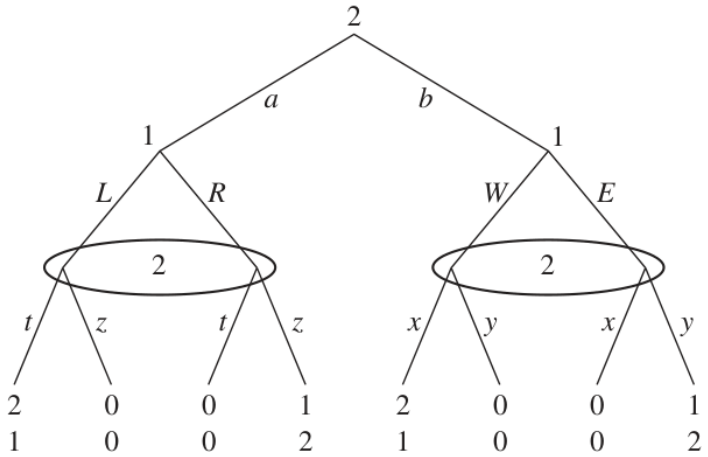


FIGURE 7.14 Exercise 7.10.

a. What are the pure-strategy sets for each player?

b. Show that for any behavioral strategy for player 1 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 2.

c. Show that for any behavioral strategy for player 2 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 1.

d. Now imagine that the game does not have perfect recall so that player 2's bottom two information sets are now one large information set. Can you find an example showing that the claim in (c) is no longer true?

(a) What is asked. In the game of Figure 7.14: (a) pure-strategy sets; (b) every behavioral strategy of player 1 has an outcome-equivalent mixed strategy; (c) the same for player 2; (d) whether (c) fails once player 2's two bottom information sets are merged.

(b) Method. Pure, mixed and behavioral strategies (Lecture 3, slides 25 and 27; Tadelis §7.2.1–7.2.2, Defs. 7.4–7.6, pp. 137–140); perfect recall and Kuhn's equivalence theorem (Lecture 3, slide 28; Tadelis Def. 7.7 and the discussion on pp. 141–142). Outcome equivalence is checked node by node: two strategies of a player are equivalent if, against every pure strategy of the opponent (and then, by linearity, every mixed one), they put the same probability on each terminal node.

(c) Solution. The game. Player 2 moves first (a or b). After a, player 1 chooses L or R; after b, he chooses W or E. Player 2 then moves again without seeing player 1's choice: at information set $h_a = \{aL, aR\}$ with actions t, z, and at $h_b = \{bW, bE\}$ with actions x, y. Payoffs (u_1, u_2): $(2, 1)$ after Lt and Wx, $(1, 2)$ after Rz and Ey, $(0, 0)$ otherwise.

(a) Pure strategies. Player 1 has two information sets (after a, after b), each with two actions:

$$S_1 = \{L, R\} \times \{W, E\} = \{LW, LE, RW, RE\} \quad (4).$$

Player 2 has three information sets (the root, h_a, h_b):

$$S_2 = \{a, b\} \times \{t, z\} \times \{x, y\} \quad (2^3 = 8),$$

e.g. (a, t, x), (a, t, y), $\ldots$, (b, z, y).

(b) Player 1: behavioral $\Rightarrow$ mixed. A behavioral strategy is $\beta_1 = (\lambda, \omega)$, with $\lambda = \Pr(L)$ at the left node and $\omega = \Pr(W)$ at the right node. Define the mixed strategy as the product

$$\sigma_1(LW) = \lambda\omega, \quad \sigma_1(LE) = \lambda(1-\omega), \quad \sigma_1(RW) = (1-\lambda)\omega, \quad \sigma_1(RE) = (1-\lambda)(1-\omega).$$

Fix any pure $s_2 = (r, m_a, m_b)$. If $r = a$, the play reaches player 1's left node and ends at (a, L, m_a) or (a, R, m_a). Under β_1 these have probabilities λ and $1-\lambda$. Under σ_1, (a, L, m_a) occurs iff the drawn plan starts with L:

$$\sigma_1(LW) + \sigma_1(LE) = \lambda\omega + \lambda(1-\omega) = \lambda,$$

and (a, R, m_a) has probability $1-\lambda$. The case $r = b$ is symmetric with ω. All other terminal nodes have probability 0 under both. So β_1 and σ_1 induce the same distribution against every pure s_2, hence (expected values are linear in σ_2) against every mixed σ_2 too. Since player 1 is called once per play, only the marginals matter: any σ_1 with $\sigma_1(LW) = \mu$, $\sigma_1(LE) = \lambda - \mu$, $\sigma_1(RW) = \omega - \mu$, $\sigma_1(RE) = 1 - \lambda - \omega + \mu$, $\mu \in [\max\{0, \lambda + \omega - 1\}, \min\{\lambda, \omega\}]$, also works.

(c) Player 2: behavioral $\Rightarrow$ mixed. A behavioral strategy is $\beta_2 = (\alpha, \tau, \xi)$: $\Pr(a)$ at the root, $\Pr(t)$ at h_a, $\Pr(x)$ at h_b. Define the product mixed strategy

$$\sigma_2(r, m_a, m_b) = \Pr_\alpha(r) \cdot \Pr_\tau(m_a) \cdot \Pr_\xi(m_b),$$

e.g. $\sigma_2(a, t, x) = \alpha\tau\xi$. Fix any pure $s_1 = (\ell, m)$ with $\ell \in \{L, R\}$, $m \in \{W, E\}$. Under β_2, terminal node (a, ℓ, t) has probability $\alpha\tau$. Under σ_2 it occurs iff the drawn plan has $r = a$ and $m_a = t$ (any m_b):

$$\sum_{m_b \in \{x,y\}} \sigma_2(a, t, m_b) = \alpha\tau\xi + \alpha\tau(1-\xi) = \alpha\tau.$$

The same computation gives $\alpha(1-\tau)$ for (a, ℓ, z), $(1-\alpha)\xi$ for (b, m, x) and $(1-\alpha)(1-\xi)$ for (b, m, y); nodes off s_1's choices have probability 0 under both. So the two strategies are outcome-equivalent against every s_1, hence against every σ_1. What makes this work is *perfect recall*: at h_a player 2 remembers she chose a, so her bottom move can be drawn independently of, and after, her root move, and the product of independent draws reproduces it.

(d) Imperfect recall. Merge h_a and h_b into one information set h. Actions at the nodes of an information set must be the same, so identify $x \equiv t$ and $y \equiv z$. Player 2 now has $S_2' = \{a, b\} \times \{t, z\}$ (4 pure strategies); a behavioral strategy is (α, τ); she has *forgotten* her own first move.

The claim of (c) survives. The product mixed strategy $\sigma_2(r, m) = \Pr_\alpha(r) \Pr_\tau(m)$ still reproduces (α, τ): each play passes through h at most once, so the probability of a terminal node (r, ℓ, m) is $\Pr_\alpha(r) \Pr_\tau(m)$ under both. No example can break (c) in this game. (Behavioral $\Rightarrow$ mixed fails only when one play can cross the same information set twice, as in the absent-minded driver, Tadelis p. 142.)

What fails is the converse (mixed ⇒ behavioral, the other half of Kuhn's theorem). Take

$$\sigma_2 = \tfrac{1}{2}\,(a,t) + \tfrac{1}{2}\,(b,z)$$

("go left and then play t, or go right and then play z": the plan coordinates the bottom move with the root move she no longer remembers). Against $s_1 = LW$ it gives $\Pr(a,L,t) = \frac{1}{2}$, $\Pr(b,W,z) = \frac{1}{2}$, $\Pr(a,L,z) = 0$. A behavioral (α,τ) gives

$$\Pr(a,L,t) = \alpha\tau, \qquad \Pr(a,L,z) = \alpha(1-\tau), \qquad \Pr(b,W,z) = (1-\alpha)(1-\tau).$$

$\Pr(a,L,z) = 0$ forces $\alpha = 0$ or $\tau = 1$; the first makes $\Pr(a,L,t) = 0$, the second makes $\Pr(b,W,z) = 0$, contradicting $\frac{1}{2}$ in either case. So this mixed strategy has no outcome-equivalent behavioral strategy. It also matters for payoffs: against LW, σ_2 gives player 2 $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 0 = \frac{1}{2}$, and against RE it gives $\frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 2 = 1$; behavioral strategies cannot replicate this coordination.

(d) Interpretation. Mixed strategies are plans chosen before the game; behavioral strategies are choices made at each information set. With perfect recall the two are interchangeable, which is why the course (and Tadelis) can analyze dynamic games with behavioral strategies and define SPE with them (Lecture 3, slide 34). Without recall a player who randomizes "in the moment" cannot correlate her later move with an earlier one she has forgotten, while a plan made in advance can.

(e) Verification. `check_B_L03.py` computes with sympy, for each pure strategy of the opponent and each of the eight terminal nodes, the probability under the behavioral strategy and under the product mixed strategy, for player 1 (b) and player 2 (c); all differences simplify to 0. For (d) it confirms the product construction still works and that the system $\alpha\tau = \frac{1}{2}$, $\alpha(1-\tau) = 0$, $(1-\alpha)(1-\tau) = \frac{1}{2}$ has no solution. Output line: `tadelis-7-10: (b), (c) equivalences hold symbolically; (d) mixed->behavioral fails`. Tadelis prints no answer. **Note on the statement:** read literally, (d) asks for a failure of behavioral ⇒ mixed, which cannot fail here (each play crosses the merged set once); the example above shows the failure of the converse, which is the equivalence that perfect recall protects in this game.

(f) Common mistakes.

- Giving player 2 only $2 \times 2 = 4$ strategies in (a): with perfect recall she has two separate bottom information sets, so $2^3 = 8$.
- Proving (b) only against one strategy of player 2: equivalence must hold against all of them.
- In (d), claiming that the product construction breaks. It does not; the coordination of a mixed plan is what behavioral strategies lose.

Answer. (a) $S_1 = \{LW, LE, RW, RE\}$; $S_2 = \{a,b\} \times \{t,z\} \times \{x,y\}$ (8 strategies). (b), (c) The product of the behavioral probabilities is a mixed strategy with the same distribution over terminal nodes against every opponent strategy. (d) Behavioral ⇒ mixed still holds (each play crosses the merged set once); the equivalence that fails is mixed ⇒ behavioral: $\frac{1}{2}(a,t) + \frac{1}{2}(b,z)$ has no behavioral equivalent.

Item 11: Tadelis Exercise 8.5

Exam · difficulty 3/5

Statement (verbatim from the list, item 11)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.5 "Technology Adoption", p. 170 [PDF 189]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

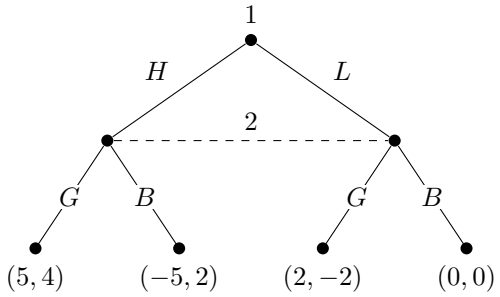
> **Technology Adoption:** During the adoption of a new technology a CEO (player 1) can design a new task for a division manager. The new task can be either high level (H) or low level (L). The manager simultaneously chooses to invest in good training (G) or bad training (B). The payoffs from this interaction are given by the following matrix:
>
> | Player 1 \ Player 2 | G | B |
> |---|---|---|
> | H | 5, 4 | -5, 2 |
> | L | 2, -2 | 0, 0 |
>
> a. Present the game in extensive form (a game tree) and solve for all the Nash equilibria and subgame-perfect equilibria.
>
> b. Now assume that before the game is played the CEO can choose not to adopt this new technology, in which case the payoffs are (1, 1), or to adopt it, in which case the game is played. Present the *entire* game in extensive form. How many proper subgames does it have?
>
> c. Solve for all the Nash equilibria and subgame-perfect equilibria of the game described in (b).

(a) What is asked. (a) Tree, Nash and subgame-perfect equilibria of the simultaneous task/training game; (b) tree and number of proper subgames once the CEO first decides whether to adopt (outside payoff $(1, 1)$); (c) Nash and subgame-perfect equilibria of that larger game.

(b) Method. Simultaneous moves as an information set (Lecture 3, slides 22–23; Tadelis Def. 7.2, p. 135); mixed equilibria by the principle of indifference (Lecture 3, slides 10–12); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); SPE by replacing the subgame with each of its Nash equilibria and optimizing at the root (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157). The structure is the voluntary battle of the sexes (Lecture 3, slide 36; Lecture 4, slide 19; Tadelis pp. 154–159).

(c) Solution. Notation. CEO = player 1 (H or L), manager = player 2 (G or B); in (b) the CEO first chooses N (no adoption) or A (adopt). $p = \Pr_1(H)$, $q = \Pr_2(G)$ in the simultaneous game.

(a) Tree and equilibria of the simultaneous game.



Pure: best replies $BR_1(G) = H$ $(5 > 2)$, $BR_1(B) = L$ $(0 > -5)$, $BR_2(H) = G$ $(4 > 2)$, $BR_2(L) = B$ $(0 > -2)$. Pure equilibria: (H, G) with $(5, 4)$ and (L, B) with $(0, 0)$.

Mixed: the manager is indifferent when

$$u_2(G) = 4p - 2(1-p) \;=\; 2p + 0 \cdot (1-p) = u_2(B) \;\Rightarrow\; 6p - 2 = 2p \;\Rightarrow\; p = \tfrac{1}{2}, \tag{3}$$

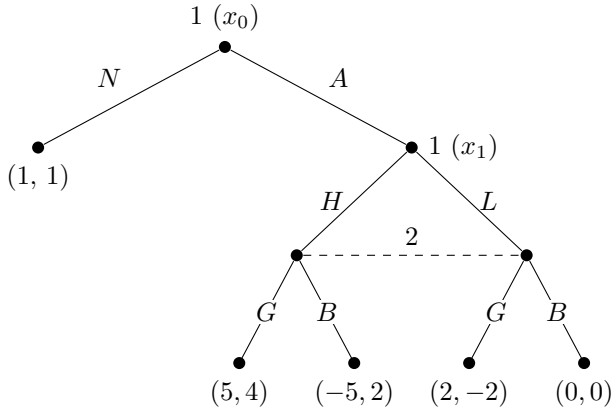
and the CEO is indifferent when

$$u_1(H) = 5q - 5(1-q) \;=\; 2q + 0 \cdot (1-q) = u_1(L) \;\Rightarrow\; 10q - 5 = 2q \;\Rightarrow\; q = \tfrac{5}{8}. \tag{4}$$

Payoffs in the mixed equilibrium: $u_1 = 2q = \frac{5}{4}$, $u_2 = 2p = 1$.

Subgame perfection: the only proper subgame is the game itself (player 2's nodes share an information set, so no subgame starts at them). Every Nash equilibrium is therefore subgame perfect: (H, G), (L, B) and $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$.

(b) The entire game. The CEO moves first at x_0: N ends the game with $(1, 1)$; A leads to x_1, where the game of (a) is played.



Candidate roots of proper subgames are singleton information sets: x_0 and x_1. Player 2's nodes are excluded (each would cut her information set). So there are **two proper subgames**: the whole game and the adoption subgame starting at x_1 (one, if the whole game is not counted, as in Gibbons).

(c) Equilibria of the entire game. The CEO now has two information sets. His strategies are $S_1 = \{NH, NL, AH, AL\}$; the manager's are $S_2 = \{G, B\}$.

	G	B
NH	$1,\ \underline{1}$	$\underline{1},\ \underline{1}$
NL	$1,\ \underline{1}$	$\underline{1},\ \underline{1}$
AH	$\underline{5},\ \underline{4}$	$-5,\ 2$
AL	$2,\ -2$	$0,\ \underline{0}$

Pure Nash equilibria: (AH, G), (NH, B), (NL, B).

All Nash equilibria. Given q, the CEO's payoffs are 1 (NH, NL), $5q - 5(1-q) = 10q - 5$ (AH) and $2q$ (AL). The manager's gain from G over B is 0 against $N\cdot$, $+2$ against AH and -2 against AL.

- $q < \frac{1}{2}$: $1 > 2q$ and $1 > 10q - 5$, so the CEO plays N (any mix of NH, NL); the manager is never reached and any q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: N and AL tie at 1 (AH gives 0). For $0 < q < 1$ the manager must be indifferent, i.e. $-2\Pr(AL) = 0$: the CEO stays on N. *Equilibria.*
- $\frac{1}{2} < q < \frac{5}{8}$: AL is the unique best reply ($2q > 1$, $2q > 10q - 5$); then the manager strictly prefers B, $q = 0$. Contradiction.
- $q = \frac{5}{8}$: AH and AL tie at $\frac{5}{4} > 1$; the manager is indifferent iff $2\Pr(AH) - 2\Pr(AL) = 0$, i.e. $\frac{1}{2}AH + \frac{1}{2}AL$. *Equilibrium.*
- $\frac{5}{8} < q \leq 1$: AH is the unique best reply; then the manager plays G, $q = 1$. *Equilibrium* (AH, G) at $q = 1$ only.

So the Nash equilibria are

$$\{(\text{any mix of } NH, NL;\ q) : q \leq \tfrac{1}{2}\} \cup \{(\tfrac{1}{2}AH + \tfrac{1}{2}AL;\ \tfrac{5}{8})\} \cup \{(AH;\ G)\}.$$

Subgame-perfect equilibria. In the adoption subgame play must be one of its three Nash equilibria; replace x_1 by its value and optimize at x_0:

Subgame equilibrium	Value (u_1, u_2)	CEO at x_0	SPE
(H, G)	$(5, 4)$	A $(5 > 1)$	(AH, G), payoffs $(5, 4)$
(L, B)	$(0, 0)$	N $(1 > 0)$	(NL, B), payoffs $(1, 1)$
$(\frac{1}{2}, \frac{5}{8})$	$(\frac{5}{4}, 1)$	A $(\frac{5}{4} > 1)$	$(A,\ \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$, payoffs $(\frac{5}{4}, 1)$

Three SPE. The Nash equilibria that are not subgame perfect are (NH, B) and every profile of the first family other than (NL, B): after adoption they prescribe (H, B) or (L, q) with $q > 0$, or a mix of H and L that is not $\frac{1}{2}$–$\frac{1}{2}$ with $q \leq \frac{1}{2}$, none of which is a Nash equilibrium of the subgame. For instance, in (NH, B) the CEO's plan H against B gives him $-5 < 0$.

(d) Interpretation. Whether a new technology is adopted depends on what each side expects the other to do after adoption, not only on its potential $(5, 4)$. If the CEO expects the manager to train badly he does not adopt, and the expectation is self-fulfilling. Adoption has an option value only if the subgame is expected to coordinate on (H, G), or at least on the mixed play, which gives $\frac{5}{4} > 1$. The outside option $(1, 1)$ works like the UK's no-deal option in the Brexit case (Lecture 4, slide 11): it sets the bar that the continuation must clear, and it eliminates the (H, B) plan, which no rational CEO would carry out after adopting.

(e) Verification. `check_B_L03.py` finds the pure equilibria of the subgame, solves (3)–(4) with sympy, checks the mixed profile and its payoffs $(\frac{5}{4}, 1)$ exactly, finds the pure equilibria (NH, B), (NL, B), (AH, G) of the 4×2 matrix, scans q on a grid of step $\frac{1}{160}$ plus $\frac{1}{2}, \frac{3}{5}, \frac{5}{8}$ (equilibria exactly at $q \leq \frac{1}{2}$, $q = \frac{5}{8}$, $q = 1$), and replaces the subgame by each equilibrium value to get the three SPE. Output line: `tadelis-8-5: subgame NE (H,G), (L,B), (1/2, 5/8); three SPE`. Tadelis prints no answer.

(f) Common mistakes.

- Treating (NH, B) and (NL, B) as the same equilibrium: they are different strategies, and only (NL, B) is subgame perfect.
- Forgetting the mixed equilibrium of the subgame, and with it the third SPE (adoption with mixed play).
- Saying the game of (a) has subgames at player 2's nodes; they belong to one information set.

Answer. (a) Nash (all subgame perfect, since the only subgame is the game itself): (H, G), (L, B), $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$ with payoffs $(5, 4)$, $(0, 0)$, $(\frac{5}{4}, 1)$. (b) Tree above; two proper subgames (the whole game and the post-adoption game). (c) Pure Nash: (AH, G), (NH, B), (NL, B); all Nash: N (any mix of NH, NL) with $\Pr(G) \leq \frac{1}{2}$, $(\frac{1}{2}AH + \frac{1}{2}AL, \frac{5}{8})$, (AH, G). SPE: (AH, G), (NL, B) and $(A, \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$.

Item 12: Tadelis Exercise 8.10

Exam · difficulty 3/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.10 "Playing It Safe", p. 172 [PDF 191]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

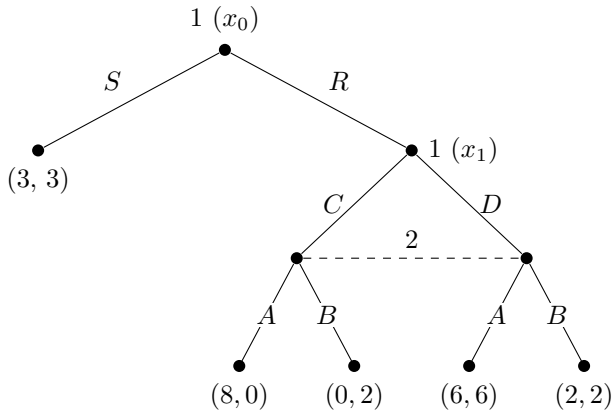
> **Playing It Safe:** Consider the following dynamic game: Player 1 can choose to play it safe (denote this choice by S), in which case both he and player 2 get a payoff of 3 each, or he can risk playing a game with player 2 (denote this choice by R). If he chooses R then they play the following simultaneous-move game:
>
> | Player 1 \ Player 2 | A | B |
> |---|---|---|
> | C | 8, 0 | 0, 2 |
> | D | 6, 6 | 2, 2 |
>
> a. Draw a game tree that represents this game. How many proper subgames does it have?
> b. Are there other game trees that would work? Explain briefly.
> c. Construct the matrix representation of the normal form of this dynamic game.
> d. Find all the Nash and subgame-perfect equilibria of the dynamic game.

(a) What is asked. Player 1 either plays safe (S, payoffs $(3,3)$) or risks (R) a simultaneous 2×2 game with player 2. (a) Draw a tree and count its proper subgames; (b) say whether other trees would do; (c) write the normal form; (d) find all Nash and subgame-perfect equilibria.

(b) Method. Extensive form with an information set for the simultaneous stage (Lecture 3, slides 22–23; Tadelis §7.1, Def. 7.2, p. 135); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); strategies as complete plans (Lecture 3, slide 25; Def. 7.4, p. 138); SPE by replacing the subgame with each of its Nash equilibria (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157; the voluntary battle of the sexes, Lecture 4, slide 19, has the same structure). Mixed equilibria by the principle of indifference (Lecture 3, slides 10–12; Tadelis §6.1–6.2, pp. 101–114).

(c) Solution. **Notation.** After R, player 1 chooses C or D and player 2 chooses A or B without seeing it. $p = \Pr_1(C)$, $q = \Pr_2(A)$ in the subgame.

(a) Tree and subgames.



Player 2's two nodes form one information set (dashed). Candidate roots of a proper subgame are the singleton nodes x_0 and x_1; player 2's nodes fail (each would cut her information set). So the game has **two proper subgames**: the whole game and the simultaneous game that starts at x_1 (one, if, like Gibbons, one does not count the whole game).

(b) Other trees. Yes. (i) In the simultaneous stage, player 2 can be drawn as moving first at x_1 with player 1's two nodes joined in an information set; the strategies, normal form and subgames are unchanged. (ii) A different tree also represents the story: player 1 picks S, RC or RD at the root, and player 2 has one information set over the nodes after RC and RD. Its normal form is the same up to merging SC and SD (it is the *reduced* normal form), but it has only one proper subgame, the whole game, so subgame perfection there has no bite: every Nash equilibrium is subgame perfect. The choice of tree matters for SPE whenever it changes which subgames exist; tree (i) and the drawing in (a) are the faithful ones, since R is a real decision taken before the stage game.

(c) Normal form. Player 1 has two information sets, so $S_1 = \{SC, SD, RC, RD\}$ (first letter at x_0, second at x_1); $S_2 = \{A, B\}$:

	A	B
SC	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
SD	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
RC	$\underline{8}$, 0	0, $\underline{2}$
RD	6, $\underline{6}$	2, 2

(d) Equilibria.

Step 1: the subgame at x_1. Best replies: $BR_1(A) = C$ ($8 > 6$), $BR_1(B) = D$ ($2 > 0$), $BR_2(C) = B$ ($2 > 0$), $BR_2(D) = A$ ($6 > 2$). They cycle, so there is no pure equilibrium. Player 2 is indifferent when

$$u_2(A) = 0 \cdot p + 6(1-p) \;=\; 2p + 2(1-p) = u_2(B) \;\Rightarrow\; 6 - 6p = 2 \;\Rightarrow\; p = \tfrac{2}{3}, \tag{5}$$

and player 1 is indifferent when

$$u_1(C) = 8q \;=\; 6q + 2(1-q) = u_1(D) \;\Rightarrow\; 8q = 2 + 4q \;\Rightarrow\; q = \tfrac{1}{2}. \tag{6}$$

The unique equilibrium of the subgame is $(p, q) = (\frac{2}{3}, \frac{1}{2})$, with values

$$u_1 = 8\left(\tfrac{1}{2}\right) = 4, \qquad u_2 = 6\left(1 - \tfrac{2}{3}\right) = 2.$$

Step 2: subgame-perfect equilibrium. Replace x_1 by $(4, 2)$. At x_0, R gives $4 > 3$, so player 1 risks. Since the subgame has a unique equilibrium, the SPE is unique:

$$\sigma^{\text{SPE}}: \text{ player 1 plays } R \text{ and then } C \text{ w.p. } \tfrac{2}{3}; \text{ player 2 plays } A \text{ w.p. } \tfrac{1}{2}, \qquad \text{payoffs } (4, 2).$$

In the normal form this is player 1 mixing $\frac{2}{3}RC + \frac{1}{3}RD$ against $\frac{1}{2}A + \frac{1}{2}B$. There is *no* pure-strategy SPE.

Step 3: all Nash equilibria. Expected payoffs against q:

Player 1's strategy	u_1 given q	player 2's gain $u_2(A) - u_2(B)$ against it
SC, SD	3	0
RC	$8q$	$0 - 2 = -2$
RD	$6q + 2(1-q) = 2 + 4q$	$6 - 2 = +4$

- $q \le \frac{1}{4}$: $3 \ge 2 + 4q$ and $3 > 8q$, so S is a best reply (strictly, for $q < \frac{1}{4}$). With player 1 on $\{SC, SD\}$ in any mix, player 2 is indifferent, so any such q is a best reply. *Nash equilibria.* (At $q = \frac{1}{4}$, RD also ties at 3, but any weight on it would make A strictly better, forcing $q = 1$; so player 1 stays on S.)
- $\frac{1}{4} < q < \frac{1}{2}$: RD is the unique best reply ($2 + 4q > 3$ and $2 + 4q > 8q$); then player 2 strictly prefers A, i.e. $q = 1$. Contradiction.
- $q = \frac{1}{2}$: RC and RD tie at $4 > 3$; player 2 is indifferent only if $-2\Pr(RC) + 4\Pr(RD) = 0$, i.e. $\frac{2}{3}RC + \frac{1}{3}RD$. *The SPE above.*
- $q > \frac{1}{2}$: RC is the unique best reply; then player 2 plays B, $q = 0$. Contradiction.

So the Nash equilibria are

$$\{\,(\text{any mix of } SC, SD;\ q) : 0 \le q \le \tfrac{1}{4}\,\} \ \cup\ \{\,(\tfrac{2}{3}RC + \tfrac{1}{3}RD;\ q = \tfrac{1}{2})\,\},$$

and the *pure* Nash equilibria are (SC, B) and (SD, B), both with payoffs $(3, 3)$. None of the first family is subgame perfect: each asks player 2 to play A with probability at most $\frac{1}{4}$ in a subgame whose only equilibrium has $q = \frac{1}{2}$. The "threat" to play B deters R, but it is not credible once R has been played.

(d) Interpretation. Playing safe is supported only by player 2's threat to play B in the risky game. Once the risky game starts, both must play its (mixed) equilibrium, which gives player 1 an expected 4, more than the safe 3, so he takes the risk. Player 2 is worse off ($2 < 3$): she would like to commit to B, but cannot; a public, costly commitment (Lecture 4, slide 11: observable, sunk moves) would be her only way to keep player 1 on the safe path. The exercise also shows that subgame perfection can select a *mixed* continuation, so the credibility test applies to probabilities, not only to pure threats.

(e) Verification. `check_B_L03.py` confirms that the subgame has no pure equilibrium, solves (5)–(6) with sympy, finds the pure Nash equilibria (SC, B), (SD, B) of the 4×2 matrix, and scans q over a grid of 241 points plus the critical values $\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$: a Nash equilibrium exists exactly at $q \in [0, \frac{1}{4}]$ and $q = \frac{1}{2}$, and at $q = \frac{1}{2}$ the mix $(\frac{2}{3}, \frac{1}{3})$ on (RC, RD) passes the exact Nash test while $(\frac{1}{2}, \frac{1}{2})$ fails. Output line: `tadelis-8-10: unique SPE (R; 2/3 C; 1/2 A), payoffs (4, 2).` Tadelis prints no answer.

(f) Common mistakes.

- Looking for a pure SPE: the subgame has none, so the SPE must use its mixed equilibrium.
- Swapping the indifference conditions: p (player 1's mix) is pinned down by *player 2's* indifference, and vice versa.

- Counting a subgame at each of player 2's nodes, or forgetting that (SC, B) and (SD, B) are two distinct Nash equilibria.

Answer. (a) Tree above; two proper subgames (the whole game and the simultaneous game after R). (b) Yes: player 2 can be drawn first in the stage game; a tree with root choices $\{S, RC, RD\}$ gives the same reduced normal form but no proper subgame other than the whole game. (c) Matrix above, $S_1 = \{SC, SD, RC, RD\}$. (d) Pure Nash: (SC, B), (SD, B); all Nash: player 1 on S (any mix of SC, SD) with $\Pr(A) \leq \frac{1}{4}$, plus $(\frac{2}{3}RC + \frac{1}{3}RD,\ \frac{1}{2}A + \frac{1}{2}B)$. Unique SPE: R, then C w.p. $\frac{2}{3}$; A w.p. $\frac{1}{2}$; payoffs $(4, 2)$. No pure SPE.

Item 13: Tadelis Exercise 8.11

Exam · difficulty 3/5

Statement (verbatim from the list, item 13)

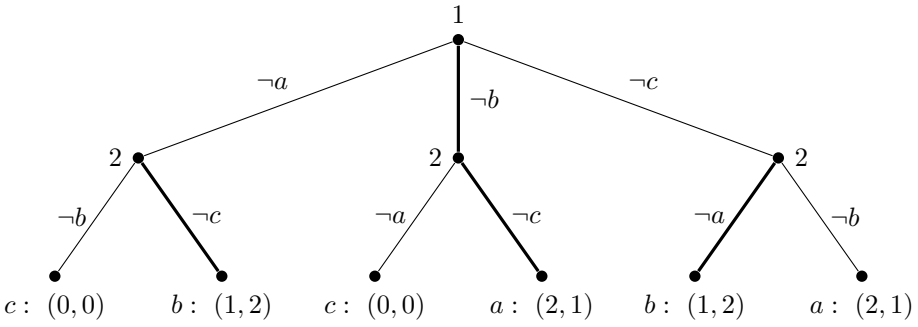
Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.11 "Resident Assistant Selection with a Twist", pp. 172–173 [PDF 191–192]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Resident Assistant Selection with a Twist:** Two staff managers in the ΠΒΦ sorority, the house manager (player 1) and kitchen manager (player 2), must select a resident assistant from a pool of three candidates: $\{a, b, c\}$. Player 1 prefers a to b, and b to c. Player 2 prefers b to a, and a to c. The process that is imposed on them is as follows: First, the house manager vetoes one of the candidates and announces the veto to the central office for staff selection and to the kitchen manager. Next the kitchen manager vetoes one of the remaining two candidates and announces it to the central office. Finally the director of the central office assigns the remaining candidate to be a resident assistant at ΠΒΦ.
>
> a. Model this as an extensive-form game (using a game tree) in which a player's most-preferred candidate gives a payoff of 2, the second most-preferred candidate gives a payoff of 1, and the last candidate gives 0.
>
> b. Find the subgame-perfect equilibrium for this game. Is it unique?
>
> c. Are there Nash equilibria that are not subgame-perfect equilibria?
>
> d. Now assume that before the two players play the game, player 2 can send an alienating e-mail to one of the candidates, which would result in that candidate withdrawing her application. Would player 2 choose to do this, and if so, with which candidate?

(a) What is asked. (a) The game tree of the two-veto procedure with payoffs 2/1/0; (b) the SPE and whether it is unique; (c) Nash equilibria that are not subgame perfect; (d) whether player 2 would use a pre-game e-mail to remove a candidate, and which one.

(b) Method. Backward induction in a finite game of perfect information (Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). Every decision node starts a subgame, so the SPE are exactly the backward-induction profiles (Def. 8.3, p. 157, and the Fact on p. 158). Nash equilibria through the normal form of the tree (§7.2.3, p. 143) and the on/off-path distinction (Def. 7.8, p. 146). Lecture 4: non-credible threats, slides 8 and 13; subgames, slide 14; SPE, slide 17.

(c) Solution. **(a) Tree.** Payoffs: $u_1(a, b, c) = (2, 1, 0)$ and $u_2(a, b, c) = (1, 2, 0)$. Player 1 vetoes one candidate, player 2 vetoes one of the two left, and the last one is assigned. "$\neg j$" means "veto j"; leaves show the assigned candidate and (u_1, u_2).



Thick branches = backward-induction choices.

Player 1 has 3 pure strategies. Player 2 has $2^3 = 8$: write $s_2 = (x_a, x_b, x_c)$, where x_j is the candidate she vetoes after player 1 vetoed j.

(b) Backward induction. Player 2's three nodes:

- after $\neg a$ ($\{b, c\}$ left): $\neg c$ leaves b, $u_2 = 2 > 0$, so $x_a = c$;
- after $\neg b$ ($\{a, c\}$ left): $\neg c$ leaves a, $u_2 = 1 > 0$, so $x_b = c$;
- after $\neg c$ ($\{a, b\}$ left): $\neg a$ leaves b, $u_2 = 2 > 1$, so $x_c = a$.

Player 1 anticipates the outcomes b, a, b of $\neg a$, $\neg b$, $\neg c$, worth $1, 2, 1$ to him, and vetoes b. The SPE is

$$(\ \neg b, \quad s_2 = (c,\ c,\ a)\ ), \qquad \text{outcome } a, \quad (u_1, u_2) = (2, 1).$$

It is unique: every comparison above is strict, so backward induction never has to break a tie. Player 1 vetoes the rival's favourite, b, not his own least preferred, c. Player 2 will veto c herself, so a veto spent on c is wasted.

(c) Nash equilibria that are not subgame perfect. Enumerating the $3 \times 8 = 24$ pure profiles gives 7 Nash equilibria:

Player 1	Player 2 (x_a, x_b, x_c)	outcome, (u_1, u_2)	SPE?
$\neg b$	(c, c, a)	a, $(2, 1)$	yes
$\neg b$	(b, c, a), (c, c, b), (b, c, b)	a, $(2, 1)$	no
$\neg a$	(c, a, a)	b, $(1, 2)$	no
$\neg c$	(c, a, a), (b, a, a)	b, $(1, 2)$	no

So yes, there are six. Two kinds:

- *Same outcome, different off-path plans.* Player 1 vetoes b and player 2 vetoes c on the path. Off the path she plans to veto b after $\neg a$ (leaving c: 0 instead of 2) or b after $\neg c$ (leaving a: 1 instead of 2). These nodes are never reached, so the plans cost her nothing, but they are not best responses in the subgames that start there.
- *A different outcome, supported by a threat.* With $x_b = a$, player 2 announces: "if you veto b, I veto a, and c gets the job." Facing it, $\neg b$ gives player 1 0, so he vetoes a or c and player 2 gets her favourite b. The threat is not credible: after $\neg b$ her choice is between a (1) and c (0), and she would veto c. Check of one of them, $(\neg c,\ (c, a, a))$: player 1 gets 1 against 0 from $\neg b$ and 1 from $\neg a$; player 2 gets 2, her maximum. The threatened node is off the path, so the threat is never tested.

No Nash equilibrium elects c: on the path, player 2 could always veto c instead and gain.

(d) The alienating e-mail. Read as follows: the e-mail comes first and its effect is seen by both. Then player 1 vetoes one of the two remaining candidates, and the last one is assigned (player 2's veto has nothing left to act on). In each two-candidate subgame player 1 keeps his preferred candidate:

Player 2's first move	candidates left	player 1 vetoes	outcome, (u_1, u_2)
no e-mail	$\{a, b, c\}$	b (part (b))	a, $(2, 1)$
e-mail a	$\{b, c\}$	c	b, $(1, 2)$
e-mail b	$\{a, c\}$	c	a, $(2, 1)$
e-mail c	$\{a, b\}$	b	a, $(2, 1)$

Player 2 gets 2 only by e-mailing a, so she does. In the SPE of the extended game, she e-mails a, player 1 vetoes c, and b is assigned, $(u_1, u_2) = (1, 2)$. The strategy profile also specifies the original SPE after "no e-mail" and player 1's vetoes after the other two e-mails.

(d) Interpretation. The order of moves decides the outcome. Moving first, player 1 removes the rival's favourite. Player 2 then cannot hurt him, because the only threat she has ("I'll leave you c") hurts her too. The e-mail works where the threat fails because it is an irreversible move made before player 1 acts, not a promise about later behaviour. Player 2 removes a, her own second choice. That looks self-defeating, but it takes away player 1's best option, and with only $\{b, c\}$ left he prefers b. A smaller choice set helps the player who controls it. This is the same commitment logic as the Brexit preparation (slide 11): an observable, sunk action changes what the opponent can do, while announcing future vetoes changes nothing ("talk is cheap", slide 11; non-credible threats, slide 13).

(e) Verification. `check_D.py` builds the tree, runs backward induction (asserting no ties), enumerates all 24 pure profiles and tests each one for Nash and for subgame perfection by definition (Nash in every subgame). It then builds the extended game with the e-mail move and solves it backward. Output: "Tadelis 8.11: SPE veto b, then (bc: veto c, ac: veto c, ab: veto a), outcome a (2,1), unique; 7 pure NE (4 with outcome a, 3 with outcome b), 6 not SPE" and "Tadelis 8.11 (d): player 2 e-mails candidate a; player 1 then vetoes c; outcome b (1,2)". (In the script, node "bc" is player 2's node after $\neg a$, and so on.) Tadelis prints no answer.

(f) Common mistakes.

- Having player 1 veto his least-preferred candidate c. Player 2 would veto c anyway; the useful veto removes her favourite.
- Writing player 2's strategy as one veto. She has three information sets, so a strategy is a triple, and it is the off-path entries that separate the Nash equilibria from the SPE.
- In (d), removing c or b ("remove the worst" or "remove the rival's favourite" by analogy). Only removing a changes player 1's choice.

Answer. (a) Tree above; player 1 has 3 strategies, player 2 has 8. (b) Unique SPE: player 1 vetoes b; player 2 vetoes c after $\neg a$, c after $\neg b$, a after $\neg c$; a is selected, payoffs $(2, 1)$. (c) Yes: 7 pure Nash equilibria, 6 not subgame perfect. Three of them elect b, payoffs $(1, 2)$, using player 2's non-credible threat to veto a after $\neg b$. (d) Yes: she e-mails a; player 1 then vetoes c and b is selected, payoffs $(1, 2)$.

Item 14: Gibbons Problem 1.14

Advanced · difficulty 4/5

Statement (verbatim from the list, item 14)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.3 Problems, Problem 1.14, p. 51 [PDF 59]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 1.14. Show that Proposition B in Appendix 1.1.C holds for mixed- as well as pure-strategy Nash equilibria: the strategies played with positive probability in a mixed-strategy Nash equilibrium survive the process of iterated elimination of strictly dominated strategies.

(a) What is asked. Show that Proposition B of Appendix 1.1.C, which says that a pure-strategy Nash equilibrium survives iterated elimination of strictly dominated strategies, holds for mixed equilibria too: every strategy played with positive probability in a mixed-strategy Nash equilibrium survives IESDS.

(b) Method. Induction on the rounds of elimination, with the principle of indifference (Lecture 3, slides 21–26) as the key step; the pure-strategy case is the same argument (Lecture 2, slide 16). Notation: $S_i^0 = S_i$ and S_i^{k+1} is the set of $s_i \in S_i^k$ that no mixed strategy $\sigma_i \in \Delta(S_i^k)$ strictly dominates on S_{-i}^k, i.e. with $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ for every $s_{-i} \in S_{-i}^k$. The survivors are $\bigcap_k S_i^k$. Dominating strategies may be mixed (Tadelis §6.3, pp. 114–117; the slides' IESDS of Lecture 1).

(c) Solution. Let $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ be a mixed-strategy Nash equilibrium and $\operatorname{supp}\sigma_i^* = \{s_i : \sigma_i^*(s_i) > 0\}$.

Claim. $\operatorname{supp}\sigma_i^* \subseteq S_i^k$ for every player i and every $k \geq 0$.

Base, $k = 0$**.** $S_i^0 = S_i$.

Step, $k \to k+1$**.** Assume $\operatorname{supp}\sigma_j^* \subseteq S_j^k$ for all j. Suppose some $s_i \in \operatorname{supp}\sigma_i^*$ is not in S_i^{k+1}. Since $s_i \in S_i^k$ by assumption, some $\sigma_i' \in \Delta(S_i^k)$ strictly dominates it on S_{-i}^k:

$$u_i(\sigma_i', s_{-i}) > u_i(s_i, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}^k.$$

By the induction hypothesis the mixed profile σ_{-i}^* puts probability one on S_{-i}^k. Averaging the strict inequality with its weights $\sigma_{-i}^*(s_{-i})$, which are non-negative and sum to one,

$$u_i(\sigma_i', \sigma_{-i}^*) \;=\; \sum_{s_{-i} \in S_{-i}^k} \sigma_{-i}^*(s_{-i})\, u_i(\sigma_i', s_{-i}) \;>\; \sum_{s_{-i} \in S_{-i}^k} \sigma_{-i}^*(s_{-i})\, u_i(s_i, s_{-i}) \;=\; u_i(s_i, \sigma_{-i}^*). \tag{7}$$

Now use equilibrium. By the principle of indifference, a strategy in the support earns the equilibrium payoff: $u_i(s_i, \sigma_{-i}^*) = u_i(\sigma^*)$. And since σ^* is a Nash equilibrium, $u_i(\sigma^*) \geq u_i(\sigma_i', \sigma_{-i}^*)$ for every σ_i'. Together,

$$u_i(s_i, \sigma_{-i}^*) = u_i(\sigma^*) \;\geq\; u_i(\sigma_i', \sigma_{-i}^*),$$

which contradicts (7). Hence every $s_i \in \operatorname{supp}\sigma_i^*$ is in S_i^{k+1}, which completes the induction. Since the claim holds for every k, the support lies in $\bigcap_k S_i^k$. ∎

Where each assumption is used. (i) Independence across players makes σ_{-i}^* a probability distribution on S_{-i}, so that the expectation (7) is a weighted average. (ii) The induction hypothesis puts that distribution inside S_{-i}^k, where the dominance inequality holds; without it the inequality

need not hold at the profiles that σ^*_{-i} uses. (iii) Indifference over the support is what turns "s_i is a best reply" into $u_i(s_i, \sigma^*_{-i}) = u_i(\sigma^*)$ for each s_i in the support, not just on average.

Why only the support. Strategies outside the support may be eliminated or not; the statement is about those played with positive probability. A strategy outside the support can be strictly dominated, and the proof says nothing about it.

Example. In matching pennies no strategy is ever dominated, so IESDS removes nothing and the support $\{H, T\}$ of the mixed equilibrium $(\frac{1}{2}, \frac{1}{2})$ survives. In the game with payoffs $u(a, L) = 3$, $u(a, R) = 0$, $u(b, L) = 0$, $u(b, R) = 3$, $u(c, \cdot) = 1$ for the row player, the strategy c is not strictly dominated by a pure strategy but is by the mix $\frac{1}{2}a + \frac{1}{2}b$ (payoff $\frac{3}{2} > 1$ against each column), so c cannot be in the support of any equilibrium.

(d) Interpretation. Mixed equilibrium puts weight only on strategies that are rationalizable by iterated dominance. So IESDS is a necessary filter for Nash equilibria in mixed as well as in pure strategies: if IESDS leaves a single profile it is the unique Nash equilibrium (the converse direction of Lecture 2, slide 17), and if a strategy is eliminated it never appears in the support of any equilibrium.

(e) Verification. `check_L03.py` draws 300 random 3×3 bimatrix games with integer payoffs, enumerates every equilibrium by support enumeration with exact fractions (checking each candidate against all pure deviations), computes the IESDS survivors with strict dominance by mixed strategies (linear programming) and asserts that every support lies inside the survivors; it also checks the example above. Output line: `gibbons-1-14: 521 NE (119 with a mixed player) in 300 random games, every support inside the IESDS survivors`. Gibbons prints no answer.

(f) Common mistakes.

- Taking the dominating strategy σ'_i to be pure: then the argument misses strategies dominated only by a mix.
- Forgetting the induction: dominance in the *original* game is not the same as dominance in the reduced game S^k, and σ^*_{-i} is known to live in S^k_{-i} only because the opponents' supports have already been shown to survive k rounds.
- Using only $u_i(\sigma^*) \geq u_i(s'_i, \sigma^*_{-i})$ for pure s'_i: the contradiction needs the equality $u_i(s_i, \sigma^*_{-i}) = u_i(\sigma^*)$ for the particular strategy s_i in the support.

Answer. Induction on k: if all supports lie in S^k, a support strategy s_i eliminated at round $k+1$ would be strictly dominated by some σ'_i on S^k_{-i}, hence against σ^*_{-i}: $u_i(\sigma'_i, \sigma^*_{-i}) > u_i(s_i, \sigma^*_{-i}) = u_i(\sigma^*)$, contradicting that σ^* is a Nash equilibrium. So all support strategies survive every round.

Item 15: Tadelis Exercise 6.3

Advanced · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 6.3 "Mixed Dominance 2", p. 123 [PDF 142]. Reproduced verbatim. **Reading, not covered in class** (Tadelis §6.3, pp. 114–117). Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

Mixed Dominance 2: Consider the game used in Section 6.3:

Player 1 \ Player 2	L	C	R
U	5, 1	1, 4	1, 0
M	3, 2	0, 0	3, 5
D	4, 3	4, 4	0, 3

a. Find a strategy different from $(\sigma_2(L), \sigma_2(C), \sigma_2(R)) = \left(0, \frac{1}{2}, \frac{1}{2}\right)$ that strictly dominates the pure strategy L for player 2. Argue that you can find an infinite number of such strategies.

b. Find a strategy different from $(\sigma_1(U), \sigma_1(M), \sigma_1(D)) = \left(0, \frac{1}{2}, \frac{1}{2}\right)$ that strictly dominates the pure strategy U for player 1 in the game remaining after one stage of elimination. Argue that you can find an infinite number of such strategies.

(a) What is asked. In the 3×3 game of Section 6.3: (a) find a mixed strategy different from $(0, \frac{1}{2}, \frac{1}{2})$ that strictly dominates L for player 2, and show that there are infinitely many; (b) the same for U of player 1 in the game that remains after one round of elimination.

(b) Method. Strict dominance by a mixed strategy: σ_i strictly dominates s_i if it earns strictly more against *every* pure strategy of the opponents (Tadelis §6.3, pp. 114–117; Gibbons Appendix 1.1.C). Take a mix of two strategies with weight a, write one inequality per opposing pure strategy and intersect the intervals for a. This is reading, not covered in the slides of Lecture 3.

(c) Solution. The game (payoffs (u_1, u_2); player 1 chooses the row):

	L	C	R
U	5, 1	1, 4	1, 0
M	3, 2	0, 0	3, 5
D	4, 3	4, 4	0, 3

(a) Dominating L. Player 2's payoffs from L against U, M, D are $1, 2, 3$; from C they are $4, 0, 4$; from R they are $0, 5, 3$. Mix C and R with $\sigma_2 = (0, a, 1-a)$. The mix must beat L against each row:

$$\begin{aligned} \text{vs } U: &\quad 4a + 0\cdot(1-a) > 1 \iff a > \tfrac{1}{4}, \\ \text{vs } M: &\quad 0\cdot a + 5(1-a) > 2 \iff a < \tfrac{3}{5}, \\ \text{vs } D: &\quad 4a + 3(1-a) = 3 + a > 3 \iff a > 0. \end{aligned}$$

The three conditions hold together exactly for $a \in (\frac{1}{4}, \frac{3}{5})$, a non-empty open interval. The book's mix is $a = \frac{1}{2}$. Another is $a = \frac{1}{3}$: payoffs $\frac{4}{3} > 1$, $\frac{10}{3} > 2$, $\frac{10}{3} > 3$. Because the interval is open and contains a continuum of points, there are infinitely many such strategies; indeed the same inequalities remain strict if a little weight is put on L itself, so the family is even larger. At the endpoints the mix only ties with L against U (at $a = \frac{1}{4}$) or M (at $a = \frac{3}{5}$), so those are not strictly dominating.

(b) Dominating U after one round. Round one removes L (by the mix of part (a)); nothing else is strictly dominated in the full game (check: against L the best mix of M, D for player 1 earns at most $4 < 5$ the payoff of U, so U is not dominated before L goes). In the game with columns $\{C, R\}$ player 1's payoffs are

	C	R
U	1	1
M	0	3
D	4	0

Mix M and D with $\sigma_1 = (0, b, 1-b)$:

$$\text{vs } C: \quad 0 \cdot b + 4(1-b) > 1 \iff b < \tfrac{3}{4},$$
$$\text{vs } R: \quad 3b + 0 \cdot (1-b) > 1 \iff b > \tfrac{1}{3}.$$

So every $b \in (\frac{1}{3}, \frac{3}{4})$ gives a strictly dominating strategy; the book's $b = \frac{1}{2}$ is one, $b = \frac{2}{5}$ another, and the interval is open, so again infinitely many. At $b = \frac{1}{3}$ the mix ties with U against R, at $b = \frac{3}{4}$ against C.

(d) Interpretation. Neither U nor L is dominated by a *pure* strategy, so a player who checks only pure strategies would conclude that no strategy can be eliminated. The mixed strategy is a hypothetical benchmark: a strategy that is beaten by an average is never a best reply to any belief, which is why mixed dominance and "never a best reply" coincide (Tadelis §6.3; MWG §8.C).

(e) Verification. `check_L03.py` confirms that no pure strategy dominates L or U (and none in round one of the full game), tests 200 values of a and b on a fine rational grid and asserts that strict dominance holds exactly on $(\frac{1}{4}, \frac{3}{5})$ and $(\frac{1}{3}, \frac{3}{4})$ (including that the endpoints fail), and that after removing L player 1's U is dominated while in the full game it is not. Output line: `tadelis-6-3: L dominated for a in (1/4,3/5); U dominated for b in (1/3,3/4)`. Tadelis prints no answer.

(f) Common mistakes.

- Testing dominance of U in the *full* game: against L player 1's payoff from U is 5, more than any mix of M and D can give, so U is dominated only after L is removed.
- Weak instead of strict inequalities: at the endpoints the mix ties against one row (or column), so the domination is weak, not strict.
- Forgetting that the dominating mix must beat L against *every* opposing pure strategy, not on average over them.

Answer. (a) $\sigma_2 = (0, a, 1-a)$ strictly dominates L for every $a \in (\frac{1}{4}, \frac{3}{5})$, e.g. $a = \frac{1}{3}$: infinitely many. (b) After L is removed, $\sigma_1 = (0, b, 1-b)$ strictly dominates U for every $b \in (\frac{1}{3}, \frac{3}{4})$, e.g. $b = \frac{2}{5}$: infinitely many.

Item 16: MWG Exercise 9.B.4

Advanced · difficulty 4/5

Statement (verbatim from the list, item 16)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 Exercises, Exercise 9.B.4ˆB, p. 302 [PDF 162]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> Prove that in a finite *zero-sum* game of perfect information, there are unique subgame perfect Nash equilibrium payoffs.

(a) What is asked. Prove that in a finite zero-sum game of perfect information all subgame-perfect Nash equilibria give the same payoffs.

(b) Method. Backward induction and Zermelo's theorem (MWG §9.B, Prop. 9.B.1, p. 272; Lecture 3, slide 30, whose corollary gives uniqueness only when no player has equal payoffs at two terminal nodes); subgames and SPNE (MWG Defs. 9.B.1–9.B.2, pp. 274–275; Lecture 3, slides 32–34); in a finite game of perfect information the SPNE are exactly the backward-induction profiles (MWG Prop. 9.B.2; Lecture 3, slide 35). The proof is an induction on subgames that tracks *values*, not strategies: ties may create many SPNE, and the point is that they cannot create different payoffs.

(c) Solution. Setting. Two players; a finite tree in which every information set is a single node; $u_1(z) + u_2(z) = 0$ at every terminal node z, so we can track $v(z) = u_1(z)$ and $u_2(z) = -v(z)$. For a decision node x, Γ_x is the subgame starting at x (every node starts one, by perfect information), and $C(x)$ is the set of its immediate successors. An SPNE may use behavioral strategies.

Claim. For every node x there is a number $V(x)$ such that *every* SPNE of Γ_x gives player 1 the payoff $V(x)$ (and player 2 $-V(x)$).

Proof by induction on the length of the longest path from x to a terminal node.

Base. If x is terminal, $V(x) = v(x)$.

Step. Let x be a decision node of player i and suppose the claim holds for every $y \in C(x)$. Take any SPNE σ of Γ_x.

(i) For each $y \in C(x)$, the restriction of σ to Γ_y is an SPNE of Γ_y (a subgame of Γ_y is a subgame of Γ_x). By the induction hypothesis it gives player 1 exactly $V(y)$. So if play moves from x to y, the continuation pays $(V(y), -V(y))$, whatever σ prescribes inside Γ_y.

(ii) The restriction of σ to Γ_x is a Nash equilibrium of Γ_x. Player i could change only his action at x, keeping the rest of σ; moving to y would give him $V(y)$ if $i = 1$, or $-V(y)$ if $i = 2$. Therefore every action played with positive probability at x must lead to a maximizer:

$$i = 1: \quad y \in \arg\max_{y' \in C(x)} V(y'), \qquad i = 2: \quad y \in \arg\max_{y' \in C(x)} \big(-V(y')\big) = \arg\min_{y' \in C(x)} V(y').$$

(iii) All the successors in the support therefore have the same value, so the payoff of σ in Γ_x is

$$V(x) = \max_{y \in C(x)} V(y) \ \text{ if } i = 1, \qquad V(x) = \min_{y \in C(x)} V(y) \ \text{ if } i = 2,$$

a number that does not depend on which SPNE σ we took.

Applying the claim at the root: every SPNE of the game gives $(V(x_0), -V(x_0))$. SPNE exist by Zermelo's theorem, so the set of SPNE payoffs is exactly $\{(V(x_0), -V(x_0))\}$. □

Where zero-sum is used. In step (iii). When player i is indifferent among several successors, they give *him* the same payoff, and in a two-player zero-sum game they then give the opponent the same payoff too. Without zero-sum, the opponent's payoff can differ across the tied successors. Counterexample: player 1 chooses $l \mapsto (0, 1)$ or $r \mapsto (0, 2)$; both l and r are SPNE, with different payoffs for player 2.

Two remarks.

- *Two players are needed.* With three players and $u_1 + u_2 + u_3 = 0$ the result fails: player 1 choosing between $(0, 1, -1)$ and $(0, -1, 1)$ has two SPNE with different payoffs for players 2 and 3. "Zero-sum" in the exercise means two-player zero-sum.
- *Chance moves.* If the tree has moves of Nature with known probabilities, define $V(x)$ at a chance node as the expected value of its successors' values; the same induction goes through. By the minimax theorem $V(x_0)$ is also the value of the game, and every Nash equilibrium, not only every SPNE, yields it.

(d) Interpretation. In a pure conflict, ties are harmless: if I am indifferent between two moves, so are you, so my tie-breaking cannot be used to help or hurt you. That is why chess-like games have a well-defined value even though the backward-induction strategies are far from unique. In the non-zero-sum games of Lecture 4 (entry, Brexit) a follower's indifference *is* strategically important: how she breaks it can change the leader's payoff (compare MWG 9.B.5, item "MWG 9.B.5", part (e)).

(e) Verification. `check_B_L03.py` draws 400 random two-player zero-sum trees (depth up to 4, two or three moves per node, integer payoffs in $\{-2, \ldots, 2\}$, fixed seed), computes the set of pure SPNE payoff vectors over all tie-breakings, and finds exactly one in each tree; 151 of the trees have more than one SPNE, so ties do occur. It also confirms the two counterexamples (non-zero-sum and three-player zero-sum) have two SPNE payoff vectors. Output line: `mwg-9-b-4: 400 random zero-sum trees, unique SPNE payoffs (151 with several SPNE)`. *Comparison with the MWG solutions manual* (Exercise 9.B.4, p. 9-2 of the manual; label read as "9.8.4" in the OCR): it argues that if a player has equal payoffs at two terminal nodes, the zero-sum property makes the other player indifferent too, so "there may exist many different ... SPNE ... but all of them will yield the same payoffs". **Agreement** on the result; the manual's argument speaks of terminal nodes only, and the induction above extends it to ties between subgame values, which is where it is actually needed.

(f) Common mistakes.

- Proving that the SPNE is unique: it need not be (ties); only the payoffs are.
- Using Zermelo's theorem's uniqueness clause, which requires no ties at all, instead of an argument that allows ties.
- Forgetting that SPNE may be mixed at tied nodes; the argument must cover any probability on the maximizers, which (ii)–(iii) do.

Answer. By induction on subgames: if every subgame after x has a unique SPNE value $V(y)$ for player 1, then at x any SPNE puts weight only on successors that maximize the mover's payoff, $\max_y V(y)$ for player 1 or $\min_y V(y)$ for player 2, so $V(x)$ is unique. At the root all SPNE pay $(V(x_0), -V(x_0))$. Zero-sum (with two players) is what makes the mover's ties irrelevant for the opponent.

Item 17: Gibbons Problem 2.1

Advanced · difficulty 5/5

Statement (verbatim from the list, item 17)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §2.6 Problems (Section 2.1), Problem 2.1, p. 130 [PDF 137]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.1. Suppose a parent and child play the following game, first analyzed by Becker (1974). First, the child takes an action, A, that produces income for the child, $I_C(A)$, and income for the parent, $I_P(A)$. (Think of $I_C(A)$ as the child's income net of any costs of the action A.) Second, the parent observes the incomes I_C and I_P and then chooses a bequest, B, to leave to the child. The child's payoff is $U(I_C + B)$; the parent's is $V(I_P - B) + kU(I_C + B)$, where $k > 0$ reflects the parent's concern for the child's well-being. Assume that: the action is a nonnegative number, $A \geq 0$; the income functions $I_C(A)$ and $I_P(A)$ are strictly concave and are maximized at $A_C > 0$ and $A_P > 0$, respectively; the bequest B can be positive or negative; and the utility functions U and V are increasing and strictly concave. Prove the "Rotten Kid" Theorem: in the backwards-induction outcome, the child chooses the action that maximizes the family's aggregate income, $I_C(A) + I_P(A)$, even though only the parent's payoff exhibits altruism.

(a) What is asked. Prove Becker's Rotten Kid theorem: when a child's action A produces incomes $I_C(A)$, $I_P(A)$ and an altruistic parent then chooses a bequest B, the backwards-induction outcome has the child choosing the A that maximizes $I_C(A) + I_P(A)$.

(b) Method. Backward induction in a two-stage game of perfect information with continuous actions (Lecture 3, slides 29–30; Gibbons §2.1.A, pp. 57–61; Tadelis §8.3.2): solve the parent's problem for every pair of incomes, then the child's problem given the parent's reply. The key step is that the parent's reply depends on the incomes only through their sum, and raises the child's consumption when the sum rises.

(c) Solution. Notation and assumptions. $A \geq 0$ the child's action; I_C, I_P strictly concave with maximizers $A_C, A_P > 0$; $B \in \mathbb{R}$ the bequest; U, V increasing and strictly concave; $k > 0$. Write $Y = I_C + I_P$ (family income) and $x = I_C + B$ (the child's consumption), so the parent consumes $I_P - B = Y - x$. As is standard for this problem, U and V are differentiable and the parent's problem has a solution.

Step 1: the parent's problem depends only on Y. Given (I_C, I_P) the parent solves

$$\max_{B \in \mathbb{R}} \; V(I_P - B) + k\,U(I_C + B).$$

Because B is unrestricted in sign, choosing B is the same as choosing $x = I_C + B \in \mathbb{R}$, and the problem becomes

$$\max_{x \in \mathbb{R}} \; W(x; Y) = V(Y - x) + k\,U(x). \tag{8}$$

W is strictly concave in x (a sum of strictly concave functions), so the maximizer $x^*(Y)$ is unique and satisfies the first-order condition

$$V'(Y - x^*) = k\,U'(x^*). \tag{9}$$

The bequest is $B^*(I_C, I_P) = x^*(I_C + I_P) - I_C$: it depends on both incomes, but the child's consumption depends only on their sum.

Step 2: x^* is strictly increasing in Y. Take $Y' > Y$ and suppose $x' = x^*(Y') \le x = x^*(Y)$. Then $Y' - x' > Y - x$. Strict concavity makes V' strictly decreasing and U' decreasing, so

$$V'(Y' - x') < V'(Y - x) \qquad \text{and} \qquad k\,U'(x') \ge k\,U'(x).$$

Applying (9) at Y and at Y':

$$V'(Y' - x') = k\,U'(x') \;\ge\; k\,U'(x) = V'(Y - x) \;>\; V'(Y' - x'),$$

a contradiction. Hence $x^*(Y') > x^*(Y)$. (With U, V twice differentiable, $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$: the parent passes part, not all, of an income gain to the child.)

Step 3: the child's problem. Anticipating B^*, the child chooses $A \ge 0$ to maximize

$$U\big(I_C(A) + B^*(I_C(A), I_P(A))\big) = U\big(x^*(I_C(A) + I_P(A))\big) = U\big(x^*(Y(A))\big).$$

$U \circ x^*$ is strictly increasing (Steps 1–2 and U increasing), so A maximizes the child's payoff if and only if it maximizes $Y(A) = I_C(A) + I_P(A)$.

Step 4: the maximizer exists and is unique. Y is strictly concave. For $A < \min\{A_C, A_P\}$ both incomes are increasing, and for $A > \max\{A_C, A_P\}$ both are decreasing, so Y attains its maximum on the compact interval $[\min\{A_C, A_P\}, \max\{A_C, A_P\}] \subset (0, \infty)$, at a unique point A^*. The backwards-induction outcome is

$$\big(A^*,\; B^*(I_C(A^*), I_P(A^*))\big), \qquad A^* = \arg\max_{A \ge 0}\, [\, I_C(A) + I_P(A)\,]. \qquad \square$$

Worked example (exact). $U = V = \ln$, $k = 2$, $I_C = 4A - A^2$ ($A_C = 2$), $I_P = 10A - A^2$ ($A_P = 5$). The parent's condition (9) reads $\frac{1}{Y - x} = \frac{2}{x}$, so $x^*(Y) = \frac{2Y}{3}$ (in general $x^* = \frac{kY}{1+k}$). The child maximizes $\ln(\frac{2}{3}(14A - 2A^2))$: $14 - 4A = 0$, $A^* = \frac{7}{2}$, between $A_C = 2$ and $A_P = 5$. There $Y = \frac{49}{2}$, $x^* = \frac{49}{3}$, $I_C = \frac{7}{4}$, so the bequest is $B^* = \frac{49}{3} - \frac{7}{4} = \frac{175}{12}$. Had the child chosen its own income maximizer $A_C = 2$, family income would be $Y(2) = 20 < \frac{49}{2}$ and its consumption only $\frac{40}{3} < \frac{49}{3}$.

(d) Interpretation. The child is selfish but faces a parent who "taxes" any loss of family income: if the child raised I_C at a larger cost to I_P, the parent, who cares about the split, would cut the bequest by more than the child gained. Altruism on one side, plus the parent's last move, aligns the first mover's incentives with joint surplus. Each assumption has a role:

- *the parent moves last*: it is the parent's credible, sequentially rational reply that disciplines the child (Lecture 4's theme: what matters is what a player will actually do when the moment comes);
- *B unrestricted*: if $B \ge 0$ were imposed and binding, the child's consumption would be I_C alone over a range and Step 1 fails;
- *the child moves first and cannot commit the parent*: if the bequest were fixed before A, the child would simply choose A_C.

(e) Verification. `check_B_L03.py` checks with sympy that $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$ for several negative (V'', U'') and positive k; derives $x^* = kY/(1 + k)$ for log utility (second derivative negative) and $x^* = (Y + \ln k)/2$ for CARA utility $-e^{-x}$; solves the child's problem in the log example ($A^* = \frac{7}{2}$), confirms on an exact grid of step $\frac{1}{20}$ that the maximizer of $\frac{2}{3}Y(A)$ is $\frac{7}{2}$ while I_C alone peaks at 2, and recomputes $B^* = \frac{175}{12}$. Output line: `gibbons-2-1: x*(Y) increasing; child picks A* = 7/2 = argmax(I_C + I_P), not A_C = 2`. Gibbons prints no answer.

(f) Common mistakes.

- Assuming the child maximizes I_C: its payoff depends on the bequest, which reacts to A.
- Treating the bequest as a fixed number instead of a function $B^*(I_C, I_P)$ (a strategy of the parent in the second stage).
- Proving only that x^* is weakly increasing, or skipping existence/uniqueness of A^*; the strict monotonicity in Step 2 is what makes "maximize $U(x^*(Y))$" equivalent to "maximize Y".

Answer. The parent's best bequest gives the child consumption $x^*(I_C + I_P)$ that solves $V'(Y - x) = kU'(x)$; it depends only on family income Y and is strictly increasing in it. So the child's payoff $U(x^*(Y(A)))$ is a strictly increasing function of $Y(A)$, and the child chooses $A^* = \arg\max_A [I_C(A) + I_P(A)]$, which exists and is unique by strict concavity.

Item 18: MWG Exercise 9.B.5

Advanced · difficulty 5/5

Statement (verbatim from the list, item 18)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 Exercises, Exercise 9.B.5ˆB (E. Maskin), p. 302 [PDF 162]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> (E. Maskin) Consider a game with two players, player 1 and player 2, in which each player i can choose an action from a finite set M_i that contains m_i actions. Player i's payoff if the action choices are (m_1, m_2) is $\phi_i(m_1, m_2)$.
>
> (a) Suppose, first, that the two players move simultaneously. How many strategies does each player have?
>
> (b) Now suppose that player 1 moves first and that player 2 observes player 1's move before choosing her move. How many strategies does each player have?
>
> (c) Suppose that the game in (b) has multiple SPNEs. Show that if this is the case, then there exist two pairs of moves (m_1, m_2) and (m_1', m_2') (where either $m_1 \neq m_1'$ or $m_2 \neq m_2'$) such that either
>
> $$\text{(i)} \quad \phi_1(m_1, m_2) = \phi_1(m_1', m_2')$$
>
> or
>
> $$\text{(ii)} \quad \phi_2(m_1, m_2) = \phi_2(m_1', m_2').$$
>
> (d) Suppose that for any two pairs of moves (m_1, m_2) and (m_1', m_2') such that $m_1 \neq m_1'$ or $m_2 \neq m_2'$, condition (ii) is violated (i.e., player 2 is never indifferent between pairs of moves). Suppose also that there exists a pure strategy Nash equilibrium in the game in (a) in which π_1 is player 1's payoff. Show that in any SPNE of the game in (b), player 1's payoff is at least π_1. Would this conclusion necessarily hold for any *Nash* equilibrium of the game in (b)?
>
> (e) Show by example that the conclusion in (d) may fail either if condition (ii) holds for some strategy pairs (m_1, m_2), (m_1', m_2') with $m_1 \neq m_1'$ or $m_2 \neq m_2'$ or if we replace the phrase *pure strategy Nash equilibrium* with the phrase *mixed strategy Nash equilibrium.*

Optional extras.

- Tadelis 7.5 (veto power, p. 148), 7.7 (roommates voting, p. 149) and 7.8 with 8.3 (Brothers, pp. 149, 170): more trees, strategy sets and SPE.

- Tadelis 7.3 (tic-tac-toe, pp. 147–148): counting information sets in a large tree; needs a spreadsheet.

- Tadelis 8.1 and 8.2 (p. 170): mixed-strategy SPE of the voluntary Battle of the Sexes (read Fig. 8.2, p. 154) and of mutually assured destruction (read §8.3.3, pp. 163–166, Fig. 8.11); 8.1 is the slides' Example 2 of Lecture 4.

- Muñoz-Garcia Ch. 4 Ex. 2 (entry-predation, p. 156), Ex. 5–6 (sequential Pareto coordination and chicken, pp. 166, 169), Ex. 10 (trust and reciprocity, p. 177), Ex. 19–20 (backward induction drills, pp. 202, 204): all with worked answers.

- Gibbons 2.2 (Samaritan's dilemma, pp. 130–131) and 2.4 (two-period contribution game, pp. 131–132): continuous-action backward induction after item 17.

(a) What is asked. For a finite two-player game played simultaneously or with player 1 moving first and observed: (a), (b) count strategies; (c) multiple SPNE imply a payoff tie; (d) without ties for player 2, moving first guarantees player 1 at least any pure Nash payoff of the simultaneous game, in every SPNE (and whether in every Nash equilibrium); (e) examples where (d) fails with ties for player 2 or with a mixed Nash equilibrium.

(b) Method. Strategies as complete plans (MWG §7.D; Lecture 3, slide 25); SPNE and backward induction in finite games of perfect information (MWG §9.B, Prop. 9.B.1 (Zermelo), p. 272, and Prop. 9.B.2; Lecture 3, slides 30 and 35; Lecture 4, slide 17). The sequential game in (b) has perfect information, so its SPNE are the backward-induction profiles; with no ties they are unique. Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13.

(c) Solution. Notation. As in the statement, m_i is both the number of actions and (from (c) on) a typical action; write $|M_i|$ for the number to avoid confusion. In the sequential game a strategy of player 2 is a function $s_2 : M_1 \to M_2$.

(a) Simultaneous moves. Each player has one information set, so player i has $|M_i| = m_i$ pure strategies (and the mixed strategies $\Delta(M_i)$, a simplex of dimension $m_i - 1$).

(b) Player 1 first, observed. Player 1 still has one information set: m_1 pure strategies. Player 2 has one information set for each of player 1's moves, each with m_2 actions, so $m_2^{m_1}$ pure strategies (the functions $M_1 \to M_2$; item "Tadelis 7.1").

(c) Multiple SPNE imply a tie. Suppose, to the contrary, that no player is indifferent between any two distinct pairs: $\phi_i(m_1, m_2) \neq \phi_i(m_1', m_2')$ for $i = 1, 2$ whenever $(m_1, m_2) \neq (m_1', m_2')$.

- *Player 2's subgames.* After each m_1, the subgame is a one-player decision; any SPNE must play a best reply there. Since $\phi_2(m_1, \cdot)$ takes m_2 distinct values, the best reply $b(m_1)$ is unique and played with probability 1. So player 2's SPNE strategy is the unique function b.
- *Player 1.* Given b, choosing m_1 yields $\phi_1(m_1, b(m_1))$. The pairs $(m_1, b(m_1))$ differ across m_1 (their first components differ), so by assumption these m_1 payoffs are distinct, and the maximizer m_1^* is unique; any SPNE puts probability 1 on it.

So the SPNE is unique, (m_1^*, b), contradicting multiplicity. Hence if there are multiple SPNE, some player has a tie: (i) or (ii) holds for two distinct pairs. (In fact more is shown: either player 2 has a tie *within* some $\phi_2(m_1, \cdot)$, or player 1 has a tie between two pairs with different m_1.) □

(d) Moving first does not hurt player 1 (in SPNE). Assume (ii) never holds: player 2 has no ties. Let $(\hat{m}_1, \hat{m}_2)$ be a pure Nash equilibrium of the simultaneous game with $\pi_1 = \phi_1(\hat{m}_1, \hat{m}_2)$.

(i) In the simultaneous game $\hat{m}_2$ is a best reply to $\hat{m}_1$: $\phi_2(\hat{m}_1, \hat{m}_2) \geq \phi_2(\hat{m}_1, m_2)$ for all m_2. With no ties, it is the *unique* best reply, so $b(\hat{m}_1) = \hat{m}_2$.

(ii) In any SPNE of the sequential game, player 2 plays $b(m_1)$ after every m_1 (with probability 1, since best replies are unique). In particular she plays $\hat{m}_2$ after $\hat{m}_1$.

(iii) An SPNE is a Nash equilibrium, so player 1's equilibrium payoff is at least what he gets by deviating to $\hat{m}_1$, which is $\phi_1(\hat{m}_1, b(\hat{m}_1)) = \phi_1(\hat{m}_1, \hat{m}_2) = \pi_1$.

So in every SPNE player 1 gets at least π_1 (for every pure Nash equilibrium of (a), hence at least the largest such π_1). □

Not for every Nash equilibrium of (b). A Nash equilibrium of the sequential game may use a non-credible threat by player 2 after moves that player 1 does not make. Example (no ties for player 2; rows player 1, columns player 2, both choose O or F):

	O	F
O	2, 1	0, 0
F	0, −1	1, 2

Player 2's payoffs $1, 0, -1, 2$ are all distinct. The simultaneous game has the pure equilibrium (O, O) with $\pi_1 = 2$ (and (F, F) with 1). In the sequential game the unique SPNE is $(O,\ s_2 = (O \text{ after } O, F \text{ after } F))$, payoff 2 to player 1, as (d) predicts. But $(F,\ s_2 = (F, F))$ is also a Nash equilibrium: against "always F", player 1 gets 0 from O and 1 from F; against F, player 2's on-path choice F is optimal ($2 > -1$). Player 1 gets $1 < 2 = \pi_1$. The threat "F after O" ($0 < 1$ for player 2) is not credible.

(e) Two ways the conclusion of (d) fails.

(e.1) Player 2 has ties. Take the game of the MWG manual, with player 2's payoff depending only on player 1's move:

	L	R
U	0, 1	1, 1
D	0, 0	1, 0

(U, R) is a pure Nash equilibrium of the simultaneous game with $\pi_1 = 1$ (player 1: $1 \geq 1$; player 2: $1 \geq 1$). In the sequential game player 2 is indifferent at both nodes, so every reply is sequentially rational, and $(U, (L \text{ after } U, L \text{ after } D))$ is an SPNE (player 1 gets 0 from either move). Player 1's SPNE payoff is $0 < 1 = \pi_1$.

(e.2) A mixed Nash equilibrium instead of a pure one. Matching pennies:

	L	R
U	1, 0	0, 1
D	0, 1	1, 0

Player 2 has no ties within either row. The simultaneous game has no pure equilibrium; its mixed equilibrium $(\frac{1}{2}, \frac{1}{2})$ gives player 1 $\frac{1}{2}$. In the sequential game player 2 replies R after U and L after D, so player 1 gets 0 whatever he does: the SPNE are $(U, (R, L))$ and $(D, (R, L))$, both with payoff $0 < \frac{1}{2}$. Moving first and being observed is a *disadvantage* when player 1 wants to be unpredictable.

(d) Interpretation. Moving first helps when it is a commitment the follower must respond to (Stackelberg, entry deterrence); (d) makes this precise: if the follower's reply to each move is pinned down, the leader can always reproduce any pure Nash outcome of the simultaneous game by playing his part of it, so he cannot lose by moving first. The two failures in (e) mark the limits: if the follower is indifferent, her tie-breaking can punish the leader at no cost to herself; and if the leader's simultaneous-move advantage comes from being unpredictable (mixing), observation destroys it. The Nash-but-not-SPNE example in (d) is the Lecture 4 point once more: threats off the path can make the leader look worse off, but they are not credible.

(e) Verification. `check_B_L04.py` enumerates all pure SPNE (a strategy $s_2 : M_1 \to M_2$ that best replies in every subgame, and a best m_1) of 3000 random games with $m_1, m_2 \in \{2, 3\}$ and integer payoffs in $\{0, \ldots, 4\}$ (fixed seed). In each of the 1795 games with several SPNE a tie (i) or (ii) is present; in each of the 166 games without ties for player 2, every SPNE gives player 1 at least the payoff of every pure Nash equilibrium of the simultaneous game. It confirms the counterexample in (d) (unique SPNE pays 2, the Nash equilibrium $(F, (F, F))$ pays 1), (e.1) (SPNE $(U, (L, L))$ pays 0 against $\pi_1 = 1$) and (e.2) (SPNE $(U, (R, L))$, $(D, (R, L))$ pay 0; mixed

value $\frac{1}{2}$). Output line: `mwg-9-b-5: (c) on 1795 random games with multiple SPNE, (d) on 166 games; examples (d), (e1), (e2) confirmed`. *Comparison with the MWG solutions manual* (Exercise 9.B.5, pp. 9-2 to 9-4; label OCR'd as "9.8.5"): (a) m_i pure strategies; (b) m_1 and $m_2^{m_1}$; (c) proof by contradiction, as here; (d) the same argument, and "low payoffs for player 1 could be sustained in a NE with incredible threats by player 2"; (e) the same two examples (the indifferent follower of Fig. 9.B.5(e.1) and matching pennies, Fig. 9.B.5(e.2)). **Agreement on every part.** One slip in the manual (or its OCR): the SPNE of the pennies game are written "$(U,(R,L))$ and $(L,(R,L))$"; the second must be $(D,(R,L))$, since L is not a move of player 1.

(f) Common mistakes.

- Giving player 2 $m_1 m_2$ strategies in (b) (pairs) instead of $m_2^{m_1}$ (functions).
- In (d), forgetting that the argument needs player 2's best reply to $\hat{m}_1$ to be *unique*: with a tie she might answer $\hat{m}_1$ differently in the sequential game, which is exactly example (e.1).
- Claiming that moving first is always an advantage: (e.2) shows it can be strictly worse.

Answer. (a) m_1 and m_2. (b) m_1 for player 1, $m_2^{m_1}$ for player 2. (c) Without ties player 2's reply to each move is unique and then player 1's best move is unique, so the SPNE is unique; multiple SPNE therefore require (i) or (ii). (d) In any SPNE player 2 answers $\hat{m}_1$ with her unique best reply $\hat{m}_2$, so player 1 can secure π_1 by playing $\hat{m}_1$; not for every Nash equilibrium (e.g. $(F,(F,F))$ in the example above gives $1 < 2$). (e) Indifferent follower: $(U,(L,L))$ is an SPNE with payoff $0 < 1$; matching pennies: SPNE payoff $0 <$ mixed Nash payoff $\frac{1}{2}$.